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Lysosomal Storage Disorder Treatments

[Clinical Policy Bulletins](#) | [Medical Clinical Policy Bulletins](#)

Number: 0442

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Brand Selection for Medically Necessary Indications for Commercial Medical Plans

Gaucher Disease

As defined in Aetna commercial policies, health care services are not medically necessary when they are more costly than alternative services that are at least as likely to produce equivalent therapeutic or diagnostic results. VPRIV is more costly to Aetna than other products for Gaucher Disease. There is a lack of reliable evidence

Policy History

[Last Review](#) 
07/01/2025
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[Review History](#) 

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that VPRIV is superior to the lower cost products for treatment of Gaucher Disease: Cerezyme and Elelyso. Therefore, Aetna considers VPRIV to be medically necessary only for members who have a contraindication, intolerance or ineffective response to both of the available equivalent alternatives: Cerezyme and Elelyso.

Pompe Disease

As defined in Aetna commercial policies, health care services are not medically necessary when they are more costly than alternative services that are at least as likely to produce equivalent therapeutic or diagnostic results. Lumizyme and Pombiliti are more costly to Aetna than other products for the treatment of late-onset Pompe Disease. There is a lack of reliable evidence that Lumizyme and Pombiliti are superior to the lower cost product for the treatment of late-onset Pompe Disease: Nexviazyme. Therefore, Aetna considers Lumizyme and Pombiliti to be medically necessary only for members who have a contraindication, intolerance or ineffective response to the available equivalent alternative: Nexviazyme.

Policy

Scope of Policy

This Clinical Policy Bulletin addresses treatments for lysosomal storage disorder for commercial medical plans. For Medicare criteria, see [Medicare Part B Criteria \(https://www.aetna.com/health-care-professionals/medicare/part-b-step.html\)](https://www.aetna.com/health-care-professionals/medicare/part-b-step.html).

Note: Requires Precertification:

Precertification of enzyme replacement drugs are required of all Aetna participating providers and members in applicable plan designs. For precertification of these drugs, call (866) 752-7021 or fax (888) 267-3277. For Statement of Medical Necessity (SMN) precertification forms, see [Specialty Pharmacy Precertification \(https://www.aetna.com/health-care-professionals/health-care-professional-forms.html\)](https://www.aetna.com/health-care-professionals/health-care-professional-forms.html).

Note: Site of Care Utilization Management Policy applies to enzyme replacement drugs (Aldurazyme, Cerezyme, Elaprase, Elelyso, Elfabrio, Fabrazyme, Kanuma, Lamzede, Lumizyme, Mepsevii, Naglazyme, Nexviazyme, Pombiliti, Vimizim, Vpriv, and Xenpozyme). For information on site of service for these drugs, see [Utilization Management Policy on Site of Care for Specialty Drug Infusions \(https://www.aetna.com/health-care-professionals/utilization-management/drug-infusion-site-of-care-policy.html\)](https://www.aetna.com/health-care-professionals/utilization-management/drug-infusion-site-of-care-policy.html).

Agalsidase beta (Fabrazyme)

I. Prescriber Specialties

This medication must be prescribed by or in consultation with a physician who specializes in the treatment of metabolic disease and/or lysosomal storage disorders.

II. Criteria for Initial Approval

Aetna considers agalsidase beta (Fabrazyme) medically necessary for treatment of Fabry disease when *all* of the following criteria are met:

- A. Member is 2 years of age or older; *and*
- B. The diagnosis of Fabry disease was confirmed by enzyme assay demonstrating a deficiency of alpha-galactosidase enzyme activity or by genetic testing, or the member is a

symptomatic obligate carrier; *and*

C. Fabrazyme will not be used in combination with Galafold.

Aetna considers all other indications as experimental, investigational, or unproven.

III. *Continuation of Therapy*

Aetna considers continuation of agalsidase beta (Fabrazyme) therapy medically necessary for members with Fabry disease who are responding to therapy (e.g., reduction in plasma globotriaosylceramide [GL-3, Gb3] or GL-3/Gb3 inclusions, improvement and/or stabilization in renal function, pain reduction).

Alglucosidase alfa (Lumizyme)

I. *Prescriber Specialties*

This medication must be prescribed by or in consultation with a physician who specializes in the treatment of metabolic disease and/or lysosomal storage disorders.

II. *Criteria for Initial Approval*

Aetna considers alglucosidase alfa (Lumizyme, formerly Myozyme) medically necessary for the treatment of Pompe disease when the diagnosis of Pompe disease was confirmed by enzyme assay demonstrating a deficiency of acid alpha-glucosidase enzyme activity or by genetic testing.

Aetna considers all other indications as experimental, investigational, or unproven.

III. *Continuation of Therapy*

Aetna considers continuation of alglucosidase alpha (Lumizyme) therapy medically necessary for members with Pompe disease who are responding to therapy (e.g., improvement, stabilization, or slowing of disease progression for motor function, walking capacity, cardiorespiratory function, decrease in left ventricular mass index (LVMI), delay in death).

Avalglucosidase alfa-ngpt (Nexviazyme)

I. Prescriber Specialties

This medication must be prescribed by or in consultation with a physician who specializes in the treatment of metabolic disease and/or lysosomal storage disorders.

II. Criteria for Initial Approval

Aetna considers avalglucosidase alfa-ngpt (Nexviazyme) medically necessary for treatment of late-onset Pompe disease when *all* of the following criteria are met:

- A. Member is 1 year of age or older; *and*
- B. Diagnosis was confirmed by enzyme assay demonstrating a deficiency of acid alpha-glucosidase enzyme activity or by genetic testing.

Aetna considers all other indications as experimental, investigational, or unproven.

III. Continuation of Therapy

Aetna considers continuation of avalglucosidase alfa-ngpt (Nexviazyme) therapy medically necessary for treatment in members requesting reauthorization for late-onset Pompe disease who are responding to therapy (e.g., improvement,

stabilization, or slowing of disease progression for motor function, walking capacity, respiratory function, or muscle strength).

Cerliponase alfa (Brineura)

I. Exclusions

Aetna considers members with *either* of the following exclusions not eligible for Brineura:

- A. Dosage of Brineura exceeds 300 mg once every other week; *or*
- B. Member has acute intraventricular access device-related complications (e.g., leakage, device failure, or device-related infection) or a ventriculoperitoneal shunt.

II. Prescriber Specialties

This medication must be prescribed by or in consultation with a physician who specializes in the treatment of metabolic disease and/or lysosomal storage disorders.

III. Criteria for Initial Approval

Aetna considers cerliponase alpha (Brineura) medically necessary for neuronal ceroid lipofuscinosis type 2 (CLN2), also known as tripeptidyl peptidase 1 (TPP1) deficiency and Jansky-Bielschowsky disease, when *both* of the following criteria are met:

- A. Diagnosis of CLN2 was confirmed by enzyme assay demonstrating a deficiency of tripeptidyl peptidase 1 (TPP1) enzyme activity or by genetic testing; *and*
- B. Cerliponase alfa will be administered by, or under the direction of a physician knowledgeable in intraventricular

administration.

Aetna considers all other indications as experimental, investigational, or unproven.

IV. *Continuation of Therapy*

Aetna considers continuation of cerliponase alfa (Brineura) therapy medically necessary for neuronal ceroid lipofuscinosis type 2 (CLN2) when *both* of the following criteria are met:

- A. The requested medication will be administered by, or under the direction of a physician knowledgeable in intraventricular administration; *and*
- B. Member has experienced no loss of ambulation or a slowed loss of ambulation from baseline.

Cipaglucosidase alfa-atga (Pombiliti)

I. *Criteria for Initial Approval*

Aetna considers cipaglucosidase alfa-atga (Pombiliti) medically necessary for treatment of late-onset Pompe disease when *all* of the following criteria are met:

- A. Member is 18 years of age or older; *and*
- B. Member weighs greater than or equal to 40 kg; *and*
- C. Diagnosis was confirmed by enzyme assay demonstrating a deficiency of acid alpha-glucosidase enzyme activity or by genetic testing; *and*
- D. Member is not improving on current enzyme replacement therapy (ERT) (e.g., Lumizyme, Nexviazyme); *and*
- E. The requested medication will be taken in combination with Opfolda (miglustat).

Aetna considers all other indications as experimental, investigational, or unproven.

II. *Continuation of Therapy*

Aetna considers continuation cipaglucosidase alfa-atge (Pombiliti) therapy medically necessary for treatment in members requesting reauthorization for late-onset Pompe disease when *both* of the following criteria are met:

- A. Member is responding to therapy (e.g., improvement, stabilization, or slowing of disease progression for motor function, walking capacity, respiratory function, or muscle strength); *and*
- B. The requested medication will be taken in combination with Opfolda (miglustat).

Elosulfase alfa (Vimizim)

I. *Prescriber Specialties*

This medication must be prescribed by or in consultation with a physician who specializes in the treatment of metabolic disease and/or lysosomal storage disorders.

II. *Criteria for Initial Approval*

Aetna considers elosulfase alfa (Vimizim) medically necessary for treatment of mucopolysaccharidosis type IVA (MPS IVA; Morquio A syndrome) when the diagnosis was confirmed by enzyme assay demonstrating a deficiency of N-acetylgalactosamine-6 sulfatase (GALNS) enzyme activity or by genetic testing.

Aetna considers all other indications as experimental, investigational, or unproven.

III. *Continuation of Therapy*

Aetna considers continuation of elosulfase alfa (Vimizim) therapy medically necessary for members requesting reauthorization for an indication listed in Section II who have a clinically positive response to therapy, which shall include improvement, stabilization, or slowing of disease progression.

Galsulfase (Naglazyme)

I. *Prescriber Specialties*

This medication must be prescribed by or in consultation with a physician who specializes in the treatment of metabolic disease and/or lysosomal storage disorders.

II. *Criteria for Initial Approval*

Aetna considers galsulfase (Naglazyme) medically necessary for the treatment of members with mucopolysaccharidosis VI (MPS VI, Maroteaux-Lamy syndrome) when the diagnosis of MPS VI was confirmed by enzyme assay demonstrating a deficiency of N-acetylgalactosamine 4-sulfatase (arylsulfatase B) enzyme activity or by genetic testing.

Aetna considers all other indications as experimental, investigational, or unproven.

III. *Continuation of Therapy*

Aetna considers continuation of galsulfase (Naglazyme) therapy medically necessary for members requesting reauthorization for an indication listed in Section II who have

a clinically positive response to therapy, which shall include improvement, stabilization, or slowing of disease progression.

Idursulfase (Elaprase)

I. Prescriber Specialties

This medication must be prescribed by or in consultation with a physician who specializes in the treatment of metabolic disease and/or lysosomal storage disorders.

II. Criteria for Initial Approval

Aetna considers idursulfase (Elaprase) medically necessary for the treatment of members with mucopolysaccharidosis II (MPS II, Hunter syndrome) when the diagnosis of MPS II was confirmed by enzyme assay demonstrating a deficiency of iduronate-2-sulfatase enzyme activity or by genetic testing.

Aetna considers all other indications as experimental, investigational, or unproven.

III. Continuation of Therapy

Aetna considers continuation of idursulfase (Elaprase) therapy medically necessary for members requesting reauthorization for an indication listed in Section II who have a clinically positive response to therapy, which shall include improvement, stabilization, or slowing of disease progression.

Imiglucerase (Cerezyme), Taliglucerase alfa (Elelyso), and Velaglucerase alfa (VPRIV)

I. Prescriber Specialties

- A. Cerezyme must be prescribed by or in consultation with physicians knowledgeable in the management of patients with Gaucher disease.
- B. Elelyso and VPRIV must be prescribed by or in consultation with a physician who specializes in the treatment of metabolic disease and/or lysosomal storage disorders.

II. *Criteria for Initial Approval*

A. Gaucher disease, type 1

Aetna considers imiglucerase (Cerezyme), taliglucerase alfa (Elelyso) or velaglucerase alfa (VPRIV) medically necessary for treatment of Gaucher disease type 1 when the diagnosis of Gaucher disease was confirmed by enzyme assay demonstrating a deficiency of beta-glucocerebrosidase (glucosidase) enzyme activity or by genetic testing.

B. Gaucher disease, type 2

Aetna considers imiglucerase (Cerezyme), taliglucerase alfa (Elelyso) or velaglucerase alfa (VPRIV) medically necessary for treatment of Gaucher disease type 2 when the diagnosis of Gaucher disease was confirmed by enzyme assay demonstrating a deficiency of beta-glucocerebrosidase (glucosidase) enzyme activity or by genetic testing.

C. Gaucher disease, type 3

Aetna considers imiglucerase (Cerezyme), taliglucerase alfa (Elelyso) or velaglucerase alfa (VPRIV) medically necessary for treatment of Gaucher disease type 3 when the diagnosis of Gaucher disease was confirmed by

enzyme assay demonstrating a deficiency of beta-glucocerebrosidase (glucosidase) enzyme activity or by genetic testing.

Aetna considers all other indications as experimental, investigational, or unproven.

III. *Continuation of Therapy*

Aetna considers continuation of imiglucerase (Cerezyme), taliglucerase alfa (Elelyso), or velaglucerase alfa (VPRIV) therapy medically necessary for treatment of an indication listed in Section I when *all* of the following criteria are met:

- A. Member meets the criteria for initial approval; *and*
- B. Member is not experiencing an inadequate response or any intolerable adverse events from therapy.

Laronidase (Aldurazyme)

I. *Prescriber Specialties*

This medication must be prescribed by or in consultation with a physician who specializes in the treatment of metabolic disease and/or lysosomal storage disorders.

II. *Criteria for Initial Approval*

Aetna considers laronidase (Aldurazyme) medically necessary for treatment of mucopolysacchararidoses I (MPS I) when *both* of the following criteria are met:

- A. Diagnosis of MPS I was confirmed by enzyme assay demonstrating a deficiency of alpha-L-iduronidase enzyme activity and/or by genetic testing; *and*

B. Member has one of the following:

1. The Hurler form (i.e., severe MPS I); *or*
2. The Hurler-Scheie form (i.e., attenuated MPS I); *or*
3. The Scheie form (Scheie syndrome: i.e., attenuated MPS I) with moderate to severe symptoms (e.g., normal intelligence, less progressive physical problems, corneal clouding, joint stiffness, valvular heart disease).

Aetna considers all other indications as experimental, investigational, or unproven.

III. *Continuation of Therapy*

Aetna considers continuation of laronidase (Aldurazyme) therapy medically necessary for members requesting reauthorization for an indication listed in Section II who have a clinically positive response to therapy, which shall include improvement, stabilization, or slowing of disease progression.

Olipudase alfa-rpcp (Xenpozyme)

I. *Prescriber Specialties*

This medication must be prescribed by or in consultation with a physician knowledgeable in the management of acid sphingomyelinase deficiency (ASMD).

II. *Criteria for Initial Approval*

Aetna considers olipudase alfa-rpcp (Xenpozyme) medically necessary for treatment of non-central nervous system manifestations of acid sphingomyelinase deficiency (ASMD) when the diagnosis is confirmed by *either* of the following:

- A. A documented deficiency of acid sphingomyelinase as measured in peripheral leukocytes, cultured fibroblasts, or lymphocytes; *or*
- B. Genetic testing results documenting a pathogenic variant(s) in the sphingomyelin phosphodiesterase-1 (*SMPD1*) gene.

Aetna considers all other indications as experimental, investigational, or unproven.

III. *Continuation of Therapy*

Aetna considers continuation of olipudase alfa-rpcp (Xenpozyme) therapy medically necessary for treatment in members requesting reauthorization for an indication listed in Section I who are responding to therapy (e.g., improvement in lung function, reduction in spleen volume, reduction in liver volume, improvement in platelet count, improvement in linear growth progression).

Pegunigalsidase alfa-iwxj (Elfabrio)

I. *Prescriber Specialties*

This medication must be prescribed by or in consultation with a physician who specializes in the treatment of metabolic disease and/or lysosomal storage disorders.

II. *Criteria for Initial Approval*

Aetna considers pegunigalsidase alfa-iwxj (Elfabrio) medically necessary for treatment of Fabry disease when *all* of the following criteria are met:

- A. Member is 18 years of age or older; *and*
- B. The diagnosis of Fabry disease was confirmed by enzyme

assay demonstrating a deficiency of alpha-galactosidase enzyme activity or by genetic testing, or the member is a symptomatic obligate carrier; *and*

- C. The requested medication will not be used in combination with Galafold.

Aetna considers all other indications as experimental, investigational, or unproven.

III. *Continuation of Therapy*

Aetna considers continuation of pegunigalsidase alfa-iwxj (Elfabrio) therapy medically necessary in members requesting reauthorization for treatment of Fabry disease who are responding to therapy (e.g., reduction in plasma globotriaosylceramide [GL-3, Gb3] or GL-3/Gb3 inclusions, improvement and/or stabilization in renal function, pain reduction).

Sebelipase alfa (Kanuma)

I. *Prescriber Specialties*

This medication must be prescribed by or in consultation with a physician who specializes in the treatment of metabolic disease and/or lysosomal storage disorders.

II. *Criteria for Initial Approval*

Aetna considers sebelipase alfa (Kanuma) medically necessary for treatment of lysosomal acid lipase (LAL) deficiency when *both* of the following criteria are met:

- A. Diagnosis of LAL deficiency was confirmed by enzyme assay demonstrating a deficiency of lysosomal acid lipase enzyme activity or by genetic testing; *and*

- B. Member has alanine aminotransferase level (ALT) greater than or equal to 1.5 times the upper limit of normal (based on the age- and gender-specific normal ranges) on two consecutive ALT measurements obtained at least one week apart.

Aetna considers all other indications as experimental, investigational, or unproven.

III. *Continuation of Therapy*

Aetna considers continuation of sebelipase alfa (Kanuma) therapy medically necessary for members with LAL deficiency who are responding to therapy (e.g., improvement, stabilization, or slowing of disease progression for weight-for-age z-score if exhibiting growth failure, low-density lipoprotein [LDL], high-density lipoprotein [HDL], triglycerides, or alanine aminotransferase [ALT]).

Velmanase alfa-tycv (Lamzede)

I. *Prescriber Specialties*

This medication must be prescribed by or in consultation with a physician who specializes in the treatment of enzyme or metabolic disorders.

II. *Criteria for Initial Approval*

Aetna considers velmanase alfa-tycv (Lamzede) medically necessary for treatment of non-central nervous system manifestations of alpha-mannosidosis when the diagnosis is confirmed by *either* of the following:

- A. A documented deficiency of alpha-mannosidase activity

- as measured in blood leukocytes or fibroblasts; *or*
- B. Genetic testing results documenting a pathogenic variant(s) in the *MAN2B1* gene.

Aetna considers all other indications as experimental, investigational, or unproven.

III. *Continuation of Therapy*

Aetna considers continuation of velmanase alfa-tycv (Lamzede) therapy medically necessary in members requesting reauthorization for an indication listed in Section I who are responding to therapy (e.g., improvement in 3-minutes stair climbing test [3MSCT] from baseline, improvement in 6-minute walking test [6MWT] from baseline, improvement in forced vital capacity [FVC, % predicted] from baseline, reduction in serum or urine oligosaccharide concentration from baseline).

Vestronidase alfa-vjbk (Mepsevii)

I. *Prescriber Specialties*

This medication must be prescribed by or in consultation with a physician who specializes in the treatment of metabolic disease and/or lysosomal storage disorders.

II. *Criteria for Initial Approval*

Aetna considers vestronidase alfa-vjbk (Mepsevii) medically necessary for mucopolysaccharidosis type VII (MPS VII, Sly Syndrome) when *both* of the following criteria are met:

- A. Diagnosis of MPS VII (MPS 7) was confirmed by enzyme assay demonstrating a deficiency of beta-glucuronidase enzyme activity or by genetic testing; *and*

B. Member has elevated urinary glycosaminoglycan (uGAG) excretion at a minimum of 2-fold over the mean normal for age at initiation of treatment with Mepsevii.

Aetna considers all other indications as experimental, investigational, or unproven.

III. *Continuation of Therapy*

Aetna considers continuation of vestronidase alfa-vjbk (Mepsevii) therapy medically necessary for members requesting reauthorization for an indication listed in Section II who have a clinically positive response to therapy, which shall include improvement, stabilization, or slowing of disease progression.

Related Policies

For Cerdelga or Zavesca, see Pharmacy Clinical Policy Bulletins (PCPBs): [Formularies & Pharmacy Clinical Policy Bulletins](https://www.aetna.com/health-care-professionals/clinical-policy-bulletins/pharmacy-clinical-policy-bulletins.html) (<https://www.aetna.com/health-care-professionals/clinical-policy-bulletins/pharmacy-clinical-policy-bulletins.html>).

See also:

- [CPB 0933 - Pegvaliase-pqpz \(Palynziq\)](https://www.aetna.com/health-care-professionals/clinical-policy-bulletins/pharmacy-clinical-policy-bulletins.html) ([./900_999/0933.html](https://www.aetna.com/health-care-professionals/clinical-policy-bulletins/pharmacy-clinical-policy-bulletins.html)).

Dosage and Administration

Note: The following information is based on highlights from the U.S. FDA-approved Prescribing Information. For dose modifications and administration instructions, please refer to the Full Prescribing Information for each product.

Aldurazyme

Hurler and Hurler-Scheie forms of Mucopolysaccharidosis I (MPS I) and for persons with the Scheie form who have moderate to severe symptoms:

The recommended dosage is 0.58 mg/kg of body weight administered once weekly as an intravenous infusion.

Source: Genzyme, 2023

Brineura

Neuronal ceroid lipofuscinosis type 2 (CLN2 disease), also known as tripeptidyl peptidase 1 (TPP1) deficiency:

- Brineura is supplied as 150 mg/5 mL (30 mg/mL) solution, two single-dose vials per carton co-packaged with Intraventricular Electrolytes Injection 5 mL in a single-dose vial;
- Brineura is administered to the cerebrospinal fluid (CSF) by infusion via a surgically implanted reservoir and catheter, administered by or under the supervision of a physician experienced in intraventricular administration;
- The recommended dosage is 300 mg administered once every other week as an intraventricular infusion followed by infusion of Intraventricular Electrolytes over approximately 4.5 hours;
- In persons less than 2 years of age, lower doses are recommended. Refer to the Full Prescribing Information for Brineura;
- Dosing is not recommended in persons less than 37 weeks post-menstrual age (gestational age at birth plus post-natal age) or those weighing less than 2.5 kg

Source: BioMarin, 2024a

Cerezyme

Gaucher disease type 1

Cerezyme (imiglucerase for injection) is administered by intravenous infusion over 1 to 2 hours. Dosage should be individualized to each person. Initial dosages range from 2.5 units/kg of body weight 3 times a week to 60 units/kg once every 2 weeks. For persons weighing 18 kg and greater, infuse the diluted Cerezyme solution over 1 to 2 hours. For persons weighing less than 18 kg, infuse the diluted Cerezyme solution over 2 hours. Titrate the dosage based on clinical manifestations of disease and therapeutic goals for the individual.

Source: Genzyme, 2022

Elaprase

Hunter syndrome (Mucopolysaccharidosis II, MPS II)

The recommended dosage is 0.5 mg per kg of body weight administered once every week as an intravenous infusion.

Source: Takeda Pharmaceuticals, 2021

Elelyso

Gaucher disease type 1

Recommended dosage in persons 4 years and older

- Treatment-naïve: 60 units/kg of body weight administered every other week as a 60 to 120 minute intravenous infusion
- Persons switching from imiglucerase: Begin Elelyso at the same unit/kg dose as the person's previous imiglucerase dose. Administer Elelyso every other week as a 60 to 120 minute intravenous infusion. Dosage adjustments can be based on achievement and maintenance of each person's

therapeutic goals.

Source: Pfizer, 2023

Elfabrio

Fabry disease

Recommended dosage is 1 mg/kg every 2 weeks administered as an intravenous infusion.

Source: Chiesi USA, 2024

Fabrazyme

Fabry disease

The recommended dosage is 1 mg/kg body weight administered every two weeks as an intravenous infusion.

Individuals who have had a positive skin test to Fabrazyme or who have tested positive for anti-Fabrazyme IgE may be rechallenged with Fabrazyme. See [Full Prescribing Information \(https://products.sanofi.us/fabrazyme/fabrazyme.pdf\)](https://products.sanofi.us/fabrazyme/fabrazyme.pdf).

Source: Genzyme, 2024

Kanuma

Lysosomal Acid Lipase (LAL) deficiency

- Infants with Rapidly Progressive LAL Deficiency Presenting within the First 6 Months of Life:
 - The recommended starting dosage is 1 mg/kg as an intravenous infusion once weekly.
 - For persons who do not achieve an optimal clinical

response, dosage is increase to 3 mg/kg once weekly.

- For persons with continued suboptimal clinical response on the 3 mg/kg once weekly dosage, dosage is further increased to 5 mg/kg once weekly.
- A suboptimal clinical response is defined as any of the following: poor growth, deteriorating biochemical markers, or persistent or worsening organomegaly.

■ Pediatric and Adult Persons with LAL Deficiency:

- The recommended dosage is 1 mg/kg as an intravenous infusion once every other week.
- For persons with a suboptimal clinical response, dosage is increased to 3 mg/kg once every other week.
- A suboptimal clinical response is defined as any of the following: poor growth, deteriorating biochemical markers [e.g., alanine aminotransferase (ALT), aspartate aminotransferase (AST)], and/or parameters of lipid metabolism [e.g., low-density lipoprotein cholesterol (LDL-c), triglycerides (TG)].

Source: Alexion, 2024

Lamzede

Treatment of non-central nervous system manifestations of alpha-mannosidosis in adults and pediatrics.

The recommended dosage of Lamzede is 1 mg/kg (actual body weight) administered once every week as an intravenous infusion.

Source: Chiesi USA, 2023

Lumizyme

Pompe disease (GAA deficiency)

The recommended dosage is 20 mg per kg body weight administered every 2 weeks as an intravenous infusion. The initial infusion rate should be no more than 1 mg/kg/hour.

Source: Genzyme, 2024

Mepsevii

Mucopolysaccharidosis VII (MPS VII, Sly syndrome)

The recommended dosage is 4 mg/kg administered every two weeks as an intravenous infusion.

Source: Ultragenyx, 2020

Naglazyme

Mucopolysaccharidosis VI (MPS VI; Maroteaux-Lamy syndrome)

The recommended dosage is 1 mg per kg of body weight administered once weekly as an intravenous infusion.

Source: BioMarin, 2024

Nexviazyme

Late-onset Pompe disease

The recommended dosage for persons weighing 30 kg or more is 20 mg/kg (of actual body weight) every two weeks administered as an intravenous infusion. The recommended dosage for persons weighing less than 30 kg is 40 mg/kg (of actual body weight) every two weeks as an intravenous infusion.

Source: Genzyme, 2023

Pombiliti

Late-onset Pompe disease

The recommended dosage of Pombiliti is 20 mg/kg (of actual body weight) administered every other week as an intravenous infusion over approximately 4 hours.

Pombiliti must be administered in combination with Opfolda. If Opfolda dose is missed, Pombiliti should not be administered. Refer to the Opfolda Prescribing Information for Opfolda dosage and administration recommendations.

Start Pombiliti in combination with Opfolda 2 weeks after the last enzyme replacement therapy (ERT) dose.

Source: Amicus Therapeutics, 2023

Vimizim*Mucopolysaccharidosis type IVA (MPS IVA; Morquio A syndrome)*

The recommended dosage is 2 mg per kg body weight administered once every week as an intravenous infusion over a minimum of 3.5 to 4.5 hours, based on infusion volume.

Source: BioMarin, 2019

VPRIV*Gaucher disease type 1*

Recommended starting dose in adults and pediatrics 4 years of age or older

- Persons naïve to Enzyme Replacement Therapy: 60 Units/kg administered every other week as a 60-minute intravenous infusion.
- Persons being treated with stable imiglucerase dosages for

Gaucher disease: Can switch to VPRIV at previous imiglucerase dose two weeks after last imiglucerase dose.

Source: Takeda Pharmaceuticals, 2021

Xenpozyme

Xenpozyme is supplied as 4 mg or 20 mg of olipudase alfa-rpcp as a lyophilized powder in a single-dose vial for reconstitution.

Acid Sphingomyelinase Deficiency (ASMD)

- Adults: Recommended starting dose is 0.1 mg/kg administered as an intravenous infusion.
- Pediatrics: Recommended starting dose is 0.03 mg/kg administered as an intravenous infusion.

See Full Prescribing Information for the recommended dose escalation and maintenance dosage, dosage modifications to reduce the risk of adverse reactions, and preparation and administration instructions.

Source: Genzyme, 2023

Experimental, Investigational, or Unproven

Aetna considers the following interventions experimental, investigational, or unproven because the safety and/or efficacy have not been established:

- Concomitant use of imiglucerase, taliglucerase alfa, and velaglucerase alfa for Gaucher disease or any other indication;
- Intrathecal idursulfase for progressive cognitive impairment in individuals with MPS II;

- In utero enzyme-replacement therapy for infantile-onset Pompe disease;
- Measurements in plasma lysosphingolipids and oxysterols for routine screening of lipid storage disorders in persons without signs and symptoms of disease;
- Measurement of anti-rhGAA antibodies as a determinant of treatment outcome in adults with late-onset Pompe disease;
- The following interventions for the treatment of mucopolysaccharidoses (not an all-inclusive list):
 - Enzyme replacement therapy with fusion proteins
 - Gene therapy
 - Hematopoietic stem cell therapy
 - Intrathecal enzyme replacement therapy
 - Metallothioneins
 - Pharmacological chaperone therapy (also known as enzyme-enhancement therapy)
 - Substrate deprivation therapy
 - Substrate reduction therapy.

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
Other CPT codes related to the CPB:	
96360 - 96361	Intravenous infusion, hydration
96365 - 96368	Intravenous infusion, for therapy, prophylaxis, or diagnosis (specify substance or drug)
96379	Unlisted therapeutic, prophylactic, or diagnostic intravenous or intra-arterial injection or infusion

Other HCPCS codes related to the CPB:

S9357	Home infusion therapy, enzyme replacement intravenous therapy; (e.g., Imiglucerase); administrative services, professional pharmacy services, care coordination, and all necessary supplies and equipment (drugs and nursing visits codes separately), per diem
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Imiglucerase (Cerezyme), taliglucerase alfa (Elelyso) and Velaglucerase alfa (VPRIV):**HCPCS codes covered if selection criteria are met:**

J1786	Injection, imiglucerase, 10 units
J3060	Injection, taliglucerase alfa, 10 units
J3385	Injection, velaglucerase alfa, 100 units

ICD-10 codes covered if selection criteria are met:

E75.22	Gaucher's disease
E75.242	Niemann-Pick disease type C

Laronidase (Aldurazyme):**HCPCS codes covered if selection criteria are met:**

J1931	Injection, laronidase, 0.1 mg
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ICD-10 codes covered if selection criteria are met:

E76.01 -	Mucopolysaccharidosis, type I [Hurler's, Hurler-Scheie and
E76.03	Scheie's syndrome] [with moderate to severe symptoms]

Olipudase alfa-rpcp (Xenpozyme):**Other CPT codes related to the CPB:**

80076	Hepatic function panel This panel must include the following: Albumin (82040) Bilirubin, total (82247) Bilirubin, direct (82248) Phosphatase, alkaline (84075) Protein, total (84155) Transferase, alanine amino (ALT) (SGPT) (84460) Transferase, aspartate amino (AST) (SGOT) (84450)
81330	SMPD1(sphingomyelin phosphodiesterase 1, acid lysosomal) (eg, Niemann-Pick disease, Type A) gene analysis, common variants (eg, R496L, L302P, fsP330)
85048	Blood count; leukocyte (WBC), automated
85049	platelet, automated
85055	Reticulated platelet assay

86355	B cells, total count
86357	Natural killer (NK) cells, total count
86359	T cells; total count
86360	absolute CD4 and CD8 count, including ratio
86361	absolute CD4 count
88233	Tissue culture for non-neoplastic disorders; skin or other solid tissue biopsy
94013	Measurement of lung volumes (ie, functional residual capacity [FRC], forced vital capacity [FVC], and expiratory reserve volume [ERV]) in an infant or child through 2 years of age
94726	Plethysmography for determination of lung volumes and, when performed, airway resistance
94727	Gas dilution or washout for determination of lung volumes and, when performed, distribution of ventilation and closing volumes
94728	Airway resistance by oscillometry
HCPCS codes covered if selection criteria are met:	
J0218	Injection, olipudase alfa-rpcp, 1 mg
ICD-10 codes covered if selection criteria are met:	
E75.240 - E75.249	Niemann-Pick disease [Acid sphingomyelinase deficiency (ASMD)]
<i>Agalsidase Beta (Fabrazyme):</i>	
HCPCS codes covered if selection criteria are met:	
J0180	Injection, agalsidase beta, 1 mg
ICD-10 codes covered if selection criteria are met:	
E75.21	Fabry (-Anderson) disease
<i>Galsulfase (Naglazyme):</i>	
HCPCS codes covered if selection criteria are met:	
J1458	Injection, galsulfase, 1 mg
ICD-10 codes covered if selection criteria are met:	
E76.29	Other mucopolysaccharidosis [MPS VI]
<i>Alglucosidase Alfa (Myozyme):</i>	

HCPCS codes covered if selection criteria are met:

J0220	Injection, alglucosidase alfa, 10 mg
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ICD-10 codes covered if selection criteria are met:

E74.02	Pompe disease [infantile onset only]
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Alglucosidase Alfa (Lumizyme) :**HCPCS codes covered if selection criteria are met:**

J0221	Injection, alglucosidase alfa, (lumizyme), 10 MG
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ICD-10 codes covered if selection criteria are met:

E74.02	Pompe disease [Pompe disease]
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Avalglucosidase alfa-ngpt (Nexviazyme):**HCPCS codes covered if selection criteria are met:**

J0219	Injection, avalglucosidase alfa-ngpt, 4 mg
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ICD-10 codes covered if selection criteria are met:

E74.02	Pompe disease [1 year of age or older]
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Idursulfase (Elaprase):**CPT codes not covered for indications listed in the CPB:**

62350 - 62351	Implantation, revision or repositioning of tunneled intrathecal or epidural catheter, for long-term medication administration via an external pump or implantable reservoir/infusion pump
62360 - 62362	Implantation or replacement of device for intrathecal or epidural drug infusion
62365	Removal of subcutaneous reservoir or pump, previously implanted for intrathecal or epidural infusion
62367 - 62370	Electronic analysis of programmable, implanted pump for intrathecal or epidural drug infusion (includes evaluation of reservoir status, alarm status, drug prescription status)
95990 - 95991	Refilling and maintenance of implantable pump or reservoir for drug delivery, spinal (intrathecal, epidural) or brain (intraventricular), includes electronic analysis of pump, when performed

HCPCS codes covered if selection criteria are met:

J1743	Injection, idursulfase, 1 mg
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ICD-10 codes covered if selection criteria are met:

E76.1	Mucopolysaccharidosis, type II [Hunter's syndrome]
Elosulfase alfa (Vimizim):	
HCPCS codes covered if selection criteria are met:	
J1322	Injection, elosulfase alfa, 1 mg
ICD-10 codes covered if selection criteria are met:	
E76.210	Morquio A mucopolysaccharidoses [MPS IVA]
Pegunigalsidase alfa-iwxj (Elfabrio):	
HCPCS codes covered if selection criteria are met:	
J2508	Injection, pegunigalsidase alfa-iwxj, 1 mg
ICD-10 codes covered if selection criteria are met:	
E75.21	Fabry (-Anderson) disease
Sebelipase alfa (Kanuma):	
Other CPT codes related to the CPB:	
82657	Enzyme activity in blood cells, cultured cells, or tissue, not elsewhere specified; nonradioactive substrate, each specimen
82658	radioactive substrate, each specimen
84460	Transferase; alanine amino (ALT) (SGPT)
HCPCS codes covered if selection criteria are met:	
J2840	Injection, sebelipase alfa, 1 mg
ICD-10 codes covered if selection criteria are met:	
E75.5	Other lipid storage disorders [Lysosomal acid lipase (LAL) enzyme deficiency]
Cerliponase alpha (Brineura):	
HCPCS codes covered if selection criteria are met:	
J0567	Injection, cerliponase alfa, 1 mg
Other CPT codes related to the CPB:	
62180 – 62258	Cerebrospinal fluid (CSF) shunt
ICD-10 codes covered if selection criteria are met:	
E75.4	Neuronal ceroid lipofuscinosis
ICD-10 codes not covered for indications listed in the CPB:	
T85.01XA – T85.09XS	Mechanical complication of ventricular intracranial (communicating) shunt

Cipaglucosidase alfa-atga (Pombiliti):**HCPCS codes covered if selection criteria are met:**

J1203	Injection, cipaglucosidase alfa-atga, 5 mg
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Other HCPCS codes related to the CPB:

G0138	Intravenous infusion of cipaglucosidase alfaatga, including provider/supplier acquisition and clinical supervision of oral administration of miglustat in preparation of receipt of cipaglucosidase alfa-atga
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J1202	Miglustat, oral, 65 mg
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ICD-10 codes covered if selection criteria are met:

E74.02	Pompe disease [late onset]
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Velmanase alfa-tycv (Lamzede):**HCPCS codes covered if selection criteria are met:**

J0217	Injection, velmanase alfa-tycv, 1 mg
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ICD-10 codes covered if selection criteria are met:

E77.1	Defects in glycoprotein degradation [alpha-mannosidosis]
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Vestronidase alfa-vjbk(Mepsevii):**HCPCS codes covered if selection criteria are met:**

J3397	Injection, vestronidase alfa-vjbk, 1 mg
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ICD-10 codes covered if selection criteria are met:

E76.29	Other mucopolysaccharidoses [type VII]
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Leukocyte and urinary glycosaminoglycan (uGAG) excretion:**Other CPT codes related to the CPB:**

85048	Blood count; leukocyte (WBC), automated [leukocyte testing]
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83864	Mucopolysaccharides, acid, quantitative [urinary glycosaminoglycan (uGAG) excretion]
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Fibroblast glucuronidase enzyme assay - no specific code:***Hematopoietic stem cell therapy - no specific code :******Metallothioneins , Substrate deprivation therapy - no specific code::*****ICD-10 codes not covered if selecton criteria are met:**

E76.01 - E76.3	Mucopolysaccharidoses
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Measurement of plasma lysophingolipids and oxysterols:***Measurement of plasma lysophingolipids and oxysterols: No***

specific code
ICD-10 codes not covered if selection criteria are met:
Z13.220 Encounter for screening for lipoid disorders [Screening lipid storage disorder]
Z13.228 Encounter for screening for other metabolic disorders [Screening lipid storage disorder]

Background

U.S. Food and Drug Administration (FDA)-Approved Indications

■ *Aldurazyme (laronidase)*

Aldurazyme is indicated for the treatment of adult and pediatric patients with Hurler and Hurler-Scheie forms of Mucopolysaccharidosis I (MPS I) and for patients with the Scheie form who have moderate to severe symptoms.

Limitations of use:

- The safety and effectiveness of treating mildly affected patients with the Scheie form have not been established.
- The effect of Aldurazyme on central nervous system manifestations of the disorder has not been determined.

■ *Brineura (cerliponase alfa)*

Brineura is indicated to slow the loss of ambulation in pediatric patients with neuronal ceroid lipofuscinosis type 2 (CLN2 disease), also known as tripeptidyl peptidase 1 (TPP1) deficiency.

- *Cerezyme (imiglucerase)*

Cerezyme is indicated for treatment of adults and pediatric patients 2 years of age and older with Type 1 Gaucher disease that results in one or more of the following conditions: anemia, thrombocytopenia, bone disease, and/or hepatomegaly or splenomegaly.

- *Elaprase (idursulfase)*

Elaprase is indicated for patients with Hunter syndrome (Mucopolysaccharidosis II, MPS II). Elaprase has been shown to improve walking capacity in patients 5 years and older. In patients 16 months to 5 years of age, no data are available to demonstrate improvement in disease-related symptoms or long term clinical outcome; however, treatment with Elaprase has reduced spleen volume similarly to that of adults and children 5 years of age and older. The safety and efficacy of Elaprase have not been established in pediatric patients less than 16 months of age.

- *Elelyso (taliglucerase alfa)*

Elelyso is indicated for the treatment of patients 4 years and older with a confirmed diagnosis of type 1 Gaucher disease.

- *Elfabrio (pegunigalsidase alfa-iwxj)*

Elfabrio is indicated for the treatment of adults with confirmed Fabry disease.

- *Fabrazyme (agalsidase beta)*

Fabrazyme is a hydrolytic lysosomal neutral glycosphingolipid-specific enzyme indicated for the treatment of adult and pediatric patients 2 years of age

and older with confirmed Fabry disease.

- *Kanuma (sebelipase alfa)*

Kanuma is a hydrolytic lysosomal cholesteryl ester and triacylglycerol-specific enzyme indicated for the treatment of patients with a diagnosis of Lysosomal Acid Lipase (LAL) deficiency.

- *Lamzede (velmanase alfa-tycv)*

Lamzede is indicated for the treatment of non-central nervous system manifestations of alpha-mannosidosis in adult and pediatric patients.

- *Lumizyme (alglucosidase alfa)*

Lumizyme is a hydrolytic lysosomal glycogen-specific enzyme indicated for patients with Pompe disease (acid alpha-glucosidase [GAA] deficiency).

- *Mepsevii (vestronidase alfa-vjbk)*

Mepsevii is indicated in pediatric and adult patients for the treatment of mucopolysaccharidosis VII (MPS VII, Sly syndrome).

Limitations of use: The effect of Mepsevii on the central nervous system manifestations of MPS VII has not been determined.

- *Naglazyme (galsulfase)*

Naglazyme is indicated for patients with mucopolysaccharidosis VI (MPS VI, Maroteaux-Lamy syndrome). Naglazyme has been shown to improve walking

and stair-climbing capacity.

- *Nexviazyme (avalglucosidase alfa-ngpt)*

Nexviazyme is indicated for the treatment of patients 1 year of age and older with late-onset Pompe disease (lysosomal acid alpha-glucosidase [GAA] deficiency).

- *Pombiliti (cipaglucosidase alfa-atga)*

Pombiliti is indicated, in combination with Opfolda, for the treatment of adult patients with late-onset Pompe disease (lysosomal acid alpha-glucosidase [GAA] deficiency) weighing greater than or equal to 40 kg and who are not improving on their current enzyme replacement therapy (ERT).

- *Vimizim (elosulfase alfa)*

Vimizim is a hydrolytic lysosomal glycosaminoglycan (GAG)-specific enzyme indicated for patients with Mucopolysaccharidosis type IVA (MPS IVA; Morquio A syndrome).

- *VPRIV (velaglucerase)*

VPRIV is a hydrolytic lysosomal glucocerebroside-specific enzyme indicated for long-term enzyme replacement therapy (ERT) for patients with type 1 Gaucher disease.

- *Xenpozyme (olipudase alfa-rpcp)*

Xenpozyme is a hydrolytic lysosomal sphingomyelin-specific enzyme indicated for treatment of non-central nervous system manifestations of acid sphingomyelinase deficiency (ASMD) in adult and pediatric patients.

Compensial Uses

- *Cerezyme (imiglucerase)*

Gaucher disease type 2 and type 3

- *Ellelyso (taliglucerase alfa)*

Gaucher disease type 2 and type 3

- *VPRIV (velaglucerase)*

Gaucher disease type 2 and type 3

Acid Sphingomyelinase Deficiency

Acid sphingomyelinase deficiency (ASMD), a lysosomal storage disease, is a rare progressive genetic disorder that results from a deficiency of the acid sphingomyelinase enzyme, which is required to metabolize lipids called sphingomyelin. Consequently, sphingomyelin and other substances accumulate in the liver, spleen, lung, and brain. Individuals with ASMD have enlarged abdomens that can cause pain, vomiting, feeding difficulties, and falls. Persons most severely affected have profound neurologic symptoms and rarely survive beyond 2 to 3 years of age. Others may survive into adulthood but die prematurely from respiratory failure (FDA, 2022; NORD, 2019; Wasserstein and Schuchman, 2021).

ASMD is highly variable and tends to occur along a continuum. Individuals with the severe early-onset form, infantile neurovisceral ASMD, were historically diagnosed with Niemann-Pick disease type A (NPD-A). The later-onset, chronic visceral form of ASMD is also referred to as Niemann-Pick disease type B (NPD-B). A phenotype with intermediate severity is also known as chronic neurovisceral ASMD (NPD-A/B). The most common presenting symptom in NPD-A is hepatosplenomegaly, usually detectable by 3

months of age; over time the liver and spleen become massive in size. Due to the high variability of the disease, some children will develop severe, life-threatening complications early in life; others will have mild disease that may go undiagnosed well into adulthood. ASMD is caused by mutations in the SMPD1 gene and is inherited in an autosomal recessive manner (Wasserstein and Schuchman, 2021).

In August 2022, the U.S. FDA approved Xenpozyme (olipudase alfa) intravenous infusion for treatment of non-central nervous system manifestations of acid sphingomyelinase deficiency (ASMD) in adult and pediatric patients. Xenpozyme provides an exogenous source of the enzyme acid sphingomyelinase (ASM), thus, it is an enzyme replacement therapy that helps reduce sphingomyelin accumulation in the liver, spleen, and lung. Xenpozyme is not expected to cross the blood-brain barrier or modulate the central nervous system (CNS) manifestations of ASMD. FDA approval was based on positive results from two clinical trials involving children (ASCEND-Peds trial) and adults (ASCEND trial).

Wasserstein et al (2022) conducted a phase 2/3, 52 week, international, double-blind, placebo-controlled trial (ASCEND; NCT02004691) to assess the efficacy and safety of olipudase alfa enzyme replacement therapy for non-central nervous system manifestations of acid sphingomyelinase deficiency (ASMD) in adults. Patients (n=36) with ASMD type A/B or type B were randomized to 1:1 to receive olipudase alfa or placebo intravenously every 2 weeks with inpatient dose escalation to 3 mg/kg. Primary efficacy endpoints were percent change from baseline to week 52 in percent predicted diffusing capacity of the lung for carbon monoxide and spleen volume (combined with splenomegaly-related score in the United States). Other outcomes included liver volume/function/sphingomyelin content, pulmonary imaging/function, platelet levels, lipid profiles, and pharmacodynamics. Least square mean percent change from

baseline to week 52 favored olipudase alfa over placebo for percent predicted diffusing capacity of the lung for carbon monoxide (22% vs 3.0% increases, $p = .0004$), spleen volume (39% decrease vs 0.5% increase, $p < .0001$), and liver volume (28% vs 1.5% decreases, $p < .0001$). Splenomegaly-related score decreased in both groups ($p = .64$). Other clinical outcomes improved in the olipudase alfa group compared with the placebo group. There were no treatment-related serious adverse events or adverse event–related discontinuations. Most adverse events were mild. The authors concluded that olipudase alfa was well tolerated and associated with significant and comprehensive improvements in disease pathology and clinically relevant endpoints compared with placebo in adults with ASMD.

Diaz et al (2021) conducted a phase 1/2, international, multicenter, open-label trial (ASCEND-Peds; NCT02292654) to assess the safety and tolerability of olipudase alfa enzyme replacement therapy for non-central nervous system manifestations of acid sphingomyelinase deficiency (ASMD) in children. Patients ($n=20$; age range 1 year to 17 years) were administered intravenously olipudase alfa once every 2 weeks (via infusion), with inpatient dose escalation to 3 mg/kg, for 64 weeks. Primary outcome was safety through week 64. Secondary outcomes included pharmacokinetics, spleen and liver volumes, lung diffusing capacity (DLco), lipid profiles, and height through week 52. All patients completed the study and continued in an extension trial. After one year of treatment (52 weeks), mean splenomegaly and hepatomegaly improved by greater than 40% ($p < 0.0001$). Mean % predicted DLco improved by 32.9% ($p = 0.0053$) in patients able to perform the test. Lipid profiles and elevated liver transaminase levels normalized. Mean height Z-scores improved by 0.56 ($p < 0.0001$). The authors concluded that olipudase alfa was generally well-tolerated with significant, comprehensive improvements in disease pathology across a range of clinically relevant endpoints in children with chronic ASMD.

Xenpozyme carries a boxed warning for severe hypersensitivity reactions, including anaphylaxis. Other labeled warnings and precautions include elevated transaminases (ALT and AST), and risk of fetal malformations during dosage initiation or escalation in pregnancy.

The most common adverse reactions in adult patients (incidence of 10% or more) are headache, cough, diarrhea, hypotension and ocular hyperemia. The most common adverse reactions in pediatric patients (incidence of 20% or more) are pyrexia, cough, diarrhea, rhinitis, abdominal pain, vomiting, headache, urticaria, nausea, rash, arthralgia, pruritus, fatigue and pharyngitis.

Alpha-Mannosidosis

Alpha-mannosidosis (AM) is a rare genetic lysosomal storage disorder in which mutations in the MAN2B1 gene inhibits the ability of the alpha-mannosidase enzyme to perform its role in breaking down complexes of sugar molecules (mannose-containing oligosaccharides). This gene mutation results in accumulation of oligosaccharides in the lysosomes which cause cells to malfunction and eventually die, leading to tissue and organ damage. Alpha-mannosidosis encompasses a continuum of signs and symptoms, ranging from mild to severe. Severe cases often result in either fetal death or death in childhood. Individuals with a milder phenotype have mild-to-moderate intellectual disability, impaired hearing, weakened immune system, distinctive facial features (e.g., a large head, prominent forehead, and protruding jaw), skeletal abnormalities, and muscle weakness. Individuals may experience primary central nervous system disease which typically includes cerebellar involvement causing ataxia. Associated medical problems can include corneal opacities, hepatosplenomegaly, aseptic destructive arthritis, and metabolic myopathy. Alpha-mannosidosis is gradually progressive in which some individuals may live into the fifth or sixth decade (FDA, 2023; NIH/NLM, 2023; Malm and Nilssen, 2019).

Velmanase alfa is a recombinant human alpha-mannosidase that has been in development for weekly intravenously administered (IV) enzyme replacement therapy (ERT) for AM. Borgwardt et al (2018) report on the results of a 52 week, phase 3, multicenter, randomized, double-blinded, placebo-controlled, parallel group trial (NCT01681953) that evaluated velmanase alfa (VA) in adult and pediatric patients with AM. Twenty-five patients (13 adults, 12 pediatrics) were randomized to weekly 1 mg/kg VA or placebo for 52 weeks. At study conclusion, placebo patients switched to VA; 23 patients continued receiving VA in compassionate-use/follow-on studies and were evaluated in the extension phase [last observation (LO)]. Co-primary endpoints were changes in serum oligosaccharide (S-oligo) and in the 3-minute stair-climb test (3MSCT). The secondary endpoints were change from baseline to week 52 in the 6-minute walk test (6MWT) and in forced vital capacity percentage (FVC %, measured by spirometry) of predicted normal value. The authors reports that the mean relative change in S-oligo in the VA arm was -77.6% [95% confidence interval (CI) -81.6 to -72.8] at week 52 and -62.9% (95% CI -85.8 to -40.0) at LO; mean relative change in the placebo arm was -24.1% (95% CI -40.3 to -3.6) at week 52 and -55.7% (95% CI -76.4 to -34.9) at LO after switch to active treatment. Mean relative change in 3MSCT at week 52 was -1.1% (95% CI -9.0 to 7.6) and - % (95% CI -13.4 to 6.5) for VA and placebo, respectively. At LO, the mean relative change was 3.9% (95% CI -5.5 to 13.2) in the VA arm and 9.0% (95% CI -10.3 to 28.3) in placebo patients after switch to active treatment. Similar improvement pattern was observed in secondary parameters. A post hoc analysis investigated whether some factors at baseline could account for treatment outcome; none of those factors were predictive of the response to VA, besides age. The authors concluded the efficacy results for the clinical endpoints assessed at 12 months, 3MSCT, 6MWT and FVC % predicted, favored the VM group and were supported by a reduction in serum oligosaccharide concentration.

Velmanase alfa (VA) was also investigated in a single arm trial in pediatric alpha-mannosidosis (AM) patients less than 6 years of age (NCT02998879). All patients had AM activity below 10% of normal at baseline. The trial enrolled five patients ranging from 3.7 to 5.9 years of age, with a mean age of 4.5 years. Patients received VA 1 mg/kg as intravenous infusion once weekly (4 patients for 24 months, 1 patient for 40 months). The mean (SD) absolute and percentage changes from baseline for serum oligosaccharides at 24 months were -7.7 (4.27) $\mu\text{mol/L}$ and -65.8% (23.1%) respectively (Chiesi USA, 2023; FDA, 2023).

In February 2023, the U.S. FDA approved velmanase alfa, brand name Lamzede (Chiesi USA, Inc.), for the treatment of non-central nervous system (CNS) manifestations of alpha-mannosidosis in adult and pediatric patients. Approval was based on outcomes from the phase 3 trial summarized by Borgwardt et al (2018) above.

Although there are no known contraindications, the most common adverse reactions (incidence greater than 20%) include hypersensitivity reactions including anaphylaxis, nasopharyngitis, pyrexia, headache, and arthralgia.

Anderson-Fabry Disease

El Dib and colleagues (2017) stated that Anderson-Fabry disease (AFD) is an X-linked recessive inborn error of glycosphingolipid metabolism caused by a deficiency of alpha-galactosidase A.

Renal failure, heart and cerebrovascular involvement reduce survival. A Cochrane review provided little evidence on the use of ERT. These investigators complemented this review through a linear regression and a pooled analysis of proportions from cohort studies. They evaluated the safety and efficacy of ERT for AFD. For the systematic review, a literature search was performed, from inception to March 2016, using Medline, Embase and LILACS.

Inclusion criteria were cohort studies, patients with AFD on ERT or

natural history, and at least 1 patient-important outcome (all-cause mortality, renal, cardiovascular or cerebrovascular events, and AEs) reported. The pooled proportion and the CI were shown for each outcome. Simple linear regressions for composite endpoints were performed. A total of 77 cohort studies involving 15,305 participants proved eligible. The pooled proportions were as follows: For renal complications, agalsidase alfa 15.3 % [95 % CI: 0.048 to 0.303; I² = 77.2 %, p = 0.0005]; agalsidase beta 6 % [95 % CI: 0.04 to 0.07; I² = not applicable]; and untreated patients 21.4 % [95 % CI: 0.1522 to 0.2835; I² = 89.6 %, p < 0.0001]. Effect differences favored agalsidase beta compared to untreated patients; for cardiovascular complications, agalsidase alfa 28 % [95 % CI: 0.07 to 0.55; I² = 96.7 %, p < 0.0001]; agalsidase beta 7 % [95 % CI: 0.05 to 0.08; I² = not applicable]; and untreated patients 26.2 % [95 % CI: 0.149 to 0.394; I² = 98.8 %, p < 0.0001]. Effect differences favored agalsidase beta compared to untreated patients; and for cerebrovascular complications, agalsidase alfa 11.1 % [95 % CI: 0.058 to 0.179; I² = 70.5 %, p = 0.0024]; agalsidase beta 3.5 % [95 % CI: 0.024 to 0.046; I² = 0 %, p = 0.4209]; and untreated patients 18.3 % [95 % CI: 0.129 to 0.245; I² = 95 % p < 0.0001]. Effect differences favored agalsidase beta over agalsidase alfa or untreated patients. A linear regression showed that Fabry patients receiving agalsidase alfa were more likely to have higher rates of composite end-points compared to those receiving agalsidase beta. The authors concluded that agalsidase beta was associated with a significantly lower incidence of renal, cardiovascular and cerebrovascular events than no ERT, and to a significantly lower incidence of cerebrovascular events than agalsidase alfa. They stated that in view of these results, the use of agalsidase beta for preventing major organ complications related to AFD can be recommended.

Fabry Disease

Fabry disease is a progressive, X-linked genetic disorder resulting from a defect in the gene for the lysosomal enzyme, alpha-GAL. This enzyme deficiency results in an accumulation of globotriosylceramide (GL-3) and other lipids in tissues throughout the body. The inability to catabolize GL-3 can lead to renal failure, cardiomyopathy, and cerebrovascular accidents. The estimated incidence of Fabry disease is 1 in 40,000 males. Diagnosis of Fabry disease is confirmed by low or absent alpha-galactosidase activity in plasma or serum, leukocytes, tears, biopsied tissues, or cultured skin fibroblasts.

Fabrazyme (agalsidase beta) is a recombinant human α -galactosidase A enzyme indicated for the treatment of Fabry disease. Agalsidase beta is internalized and transported into lysosomes where it exerts enzymatic activity and reduces accumulated GL-3.

Fabrazyme (Genzyme Corporation) was FDA approved in 2003. The safety and efficacy of Fabrazyme were assessed in four clinical studies in patients with Fabry disease and one matched analysis based on data from observational studies.

In the first clinical trial of agalsidase beta submitted to the FDA for approval, 58 Fabry patients were randomly assigned to 5 months treatment with either agalsidase beta or placebo. The primary efficacy endpoint of GL-3 inclusions in renal interstitial capillary endothelial cells, was assessed by light microscopy and was graded on an inclusion severity score ranging from 0 (normal or near normal) to 3 (severe inclusions). A GL-3 inclusion score of 0 was achieved by 20 of 29 (69 %) patients treated with agalsidase beta compared to 0 of 29 patients treated with placebo ($p < 0.001$). Similar reductions in GL-3 inclusions were observed in the capillary endothelium of the heart and skin. However, during this 5-month study, no differences between groups in symptoms or renal function were observed.

The second study was a randomized, double-blind, placebo-controlled, multinational, multicenter trial of 82 patients with Fabry disease who were naïve to enzyme replacement therapy. Of the 82 enrolled patients, 51 were randomized to the Fabrazyme group and 31 to the placebo group. Patients received either 1 mg/kg intravenous Fabrazyme or placebo every 2 weeks for up to 35 months (median follow-up 18.5 months). The primary efficacy endpoint was the time to first occurrence of a clinically significant event (renal, cardiac, or cerebrovascular event, or death). A total of 14 of 51 (28%) Fabrazyme-treated patients and 13 of 31 (42%) placebo-treated patients experienced a clinically significant event (HR 0.57, 95% CI: 0.27, 1.22).

The third study was an open-label, single-arm, multinational, multicenter pediatric study in which the safety and efficacy was established in patients (n=16) aged 8 to 16 years. All patients received Fabrazyme 1 mg/kg every two weeks for up to 48 weeks.

A long-term, observational study was conducted that assessed the rate of decline in renal function (eGFR slope) in 122 patients with Fabry disease aged 16 years and older treated with Fabrazyme. Treated patients were matched 1:1 based on age (at initiation), sex, Fabry disease subtype (classic or non-classic), and baseline eGFR to a historical cohort of untreated patients with Fabry disease. The median follow-up time was 3 years in the untreated group and 4.5 years in the treated group (maximum follow-up time 5 years in both groups). The estimated mean eGFR slope was -1.5 mL/min/1.73 m²/year in the Fabrazyme-treated group and -3.2 mL/min/1.73 m²/year in the untreated group (eGFR slope difference: 1.7 mL/min/1.73 m²/year; 95% CI: 0.5, 3.0).

In an analysis of 24 Fabrazyme-treated pediatric patients with Fabry disease aged 2 to less than 8 years at Fabrazyme initiation and with elevated plasma GL-3 levels (i.e., >7.03 µg/mL) at

baseline, plasma GL-3 levels fell within the normal range (i.e., $\leq 7.03 \mu\text{g/mL}$) in 91% (20/22), 95% (18/19), and 92% (12/13) of patients at 6, 12, and 24 months, respectively.

Boxed warning for Fabrazyme include hypersensitivity reactions, including anaphylaxis. Moreover, infusion-associated reactions may occur in which Fabrazyme infusion is to be discontinued immediately with appropriate medical treatment initiated. Thus, the manufacturer recommends that patients be administered antihistamines, antipyretics, and/or corticosteroids prior to infusion.

The most common adverse reactions (20% or more) include upper respiratory tract infection, chills, pyrexia, headache, cough, paresthesia, fatigue, peripheral edema, dizziness, and rash.

Ishii et al (2012) stated that Fabry disease is an inherited lysosomal storage disorder caused by deficient α -galactosidase A activity. Many missense mutations in Fabry disease often cause misfolded gene products, which leads to their retention in the endoplasmic reticulum by the quality control system; they are then removed by endoplasmic reticulum-associated degradation. These researchers discovered that a potent α -galactosidase A inhibitor, 1-deoxygalactonojirimycin, acts as a pharmacological chaperone to facilitate the proper folding of the mutant enzyme by binding to its active site, thereby improving its stability and trafficking to the lysosomes in mammalian cells. The oral administration of 1-deoxygalactonojirimycin to transgenic mice expressing human mutant α -galactosidase A resulted in significant increases in α -galactosidase A activity in various organs, with concomitant reductions in globotriaosylceramide, which contributes to the pathology of Fabry disease. A total of 78 missense mutations were found to be responsive to 1-deoxygalactonojirimycin. The authors concluded that these data indicated that many patients with Fabry disease could potentially benefit from pharmacological chaperone therapy.

Pisani and co-workers (2017) noted that in 2009, the agalsidase beta shortage resulted in switching to agalsidase alfa treatment for many Fabry disease patients, offering the unique opportunity to compare the effects of the 2 drugs. Because single studies describing effects of switching on the disease course were limited and inconclusive, these researchers performed a systematic review and meta-analysis of existing data. Relevant studies were identified in the PubMed, Cochrane, ISI Web, and SCOPUS databases from July 2009 to September 2015. The following parameters were analyzed: clinical events, changes in organ function or structure, disease-related symptoms, lyso-Gb3 plasma levels, and AEs. The 9 studies (217 patients) included in this systematic review showed only marginal differences in most of the evaluated parameters; 7 of these studies were included in the meta-analysis (176 patients). The pooled incidence rate of major AEs was reported for 5 studies (150 patients) and was equal to 0.04 events per person-year. No significant change was observed after the shift in glomerular filtration rate, whereas left ventricular mass index, left ventricular posterior wall dimension, and ejection fraction were significantly reduced over time. The authors concluded that these findings showed that the switch to agalsidase alfa was well-tolerated and associated with stable clinical conditions.

Sheng and colleagues (2019) noted that Fabry's disease is the most prevalent lysosomal storage disorder and is notorious for its early multi-organ involvement leading to complications, including ischemic strokes and transient ischemic attacks (TIAs). Since 2001, ERT has become the mainstay treatment for Fabry's patients but the indications are not clearly defined. In a meta-analysis, these investigators reviewed the benefit of ERT for stroke prevention in Fabry's patients. They carried out a literature search from National Center for Biotechnology information (NCBI)/PubMed database without restriction of years for systematic review purposes. A systematic review of clinical cohort studies and trials was performed with pooled analysis of proportions. The pooled

proportions and the CIs for stroke recurrence ratio were calculated for both ERT treatment group and native treatment groups. A total of 7 cohort studies and 2 RCTs involving 7,513 subjects (1,471 on ERT versus 6,042 on native treatment) met inclusion criteria. The pooled proportions analysis showed that the stroke recurrence ratio in the ERT treatment group was 8.2 % [95 % CI: 0.038 to 0.126] and in native-treatment group was 16 % [95 % CI: 0.102 to 0.217].

Effect differences favored ERT treatment group over native treatment group ($p = 0.03$). The authors concluded that this meta-analysis based on the currently available data showed that ERT for Fabry's disease has beneficial effect on stroke prevention. Female carriers and atypically affected males could be started on ERT as soon as diagnosis is made. These researchers stated that further studies are needed to support the role of ERT in stroke prevention.

In May 2023, the FDA approved Elfabrio (pegunigalsidase alfa-iwxj; Chiesi Farmaceutici S.p.A./Protalix Ltd.), a PEGylated enzyme replacement therapy (ERT), for the treatment of adults with confirmed Fabry disease. FDA approval was based on data outcomes from 2 clinical trials.

Trial 1 (NCT01678898) was an open-label dose-ranging study that evaluated Elfabrio in adults ($n=18$) diagnosed with Fabry disease who were ERT-naïve or who had not received ERT for more than 26 weeks and had a negative test for anti-pegunigalsidase alfa-iwxj IgG antibodies prior to enrollment. Two patients in the 1 mg/kg treatment group discontinued the trial after their first infusion; one of them discontinued due to severe hypersensitivity reaction. The average number of globotriaosylceramide (Gb3) inclusions per renal peritubular capillary (PTC) in renal biopsy specimens of patients was assessed by light microscopy using the quantitative Barisoni Lipid Inclusion Scoring System (BLISS). Evaluable renal biopsies were obtained at baseline and at 26 weeks of treatment in 14 of the 16 patients who completed the study. Mean change from baseline

at week 26 in the BLISS score (average number of Gb3 inclusions per renal PTC) for Elfabrio-treated patients was -3.1 (-4.8, -1.4) (95%CI) (Chiesi USA, 2023).

Trial 2 (NCT03566017) was a randomized, double-blind, and active-controlled study in ERT-experienced adults diagnosed with Fabry disease. Eligible patients were treated with agalsidase beta for at least one year prior to trial entry (the mean duration of agalsidase beta treatment prior to enrollment was 5.7 years). In the study, patients were randomly assigned to receive at least 1 dose of Elfabrio (pegunigalsidase alfa) (n=52) or agalsidase beta (n=25). The primary efficacy endpoint was the annualized rate of change in estimated glomerular filtration rate (eGFR slope) assessed over 104 weeks. The study found that Elfabrio was noninferior to agalsidase beta in controlling eGFR decline. The estimated mean eGFR slope was -2.4 and -2.3 mL/min/1.73 m²/year on Elfabrio and agalsidase beta respectively. The estimated treatment difference was -0.1 mL/min/1.73 m²/year (95% CI, -2.3, 2.1) (Chiesi USA, 2023).

The most common adverse reactions (≥15%) for Elfabrio include infusion-associated reactions, nasopharyngitis, headache, diarrhea, fatigue, nausea, back pain, pain in extremity, and sinusitis.

Gaucher Disease

Gaucher's disease, the most common of the lysosomal storage diseases, is caused by a deficiency of the enzyme glucocerebrosidase that leads to an accumulation of its substrate, glucocerebroside. Glucocerebroside can collect in the spleen, liver, kidneys, lungs, brain and bone marrow. Symptoms of the disease may include an enlarged spleen and liver, liver malfunction, skeletal disorders and bone lesions that may be painful, severe neurologic complications, swelling of lymph nodes and (occasionally) adjacent joints, distended abdomen, a brownish tint to the skin, anemia, low

blood platelets, and yellow fatty deposits on the sclera. The disease shows autosomal recessive inheritance and therefore affects both males and females.

There are three common clinical subtypes. Type 1 (non-neuronopathic) disease is the most common form of the disease and may affect members in childhood or adulthood. Depending on disease onset and severity, these members may live well into adulthood.

Type 2 (acute neuronopathic) disease typically begins within six months of birth. In addition to the usual symptoms of the disease, there is extensive and progressive brain damage in this subtype; as such, affected children typically die by age two.

Type 3 (subacute or chronic neuronopathic) disease can begin in childhood or adulthood. Members with this subtype may live into their teen years and adulthood.

Type I Gaucher disease is the most common form and affects nearly 90% of diagnosed patients. Type I Gaucher disease does not involve nervous system organs, while type II and III have nervous system involvement typically leading to mental retardation. 55-60% of type I Gaucher disease patients present before the age of 20.

There are 34 known mutations that cause Gaucher disease, but 4 genetic mutations account for 95% of cases in the Ashkenazi Jewish population and 50% of cases involving the general public. Gaucher disease occurs in about 1 out of every 75,000 births and the prevalence is equal among males and females. Amongst Ashkenazi Jew', the prevalence is about 1 in 1,000 and 1 out of every 12 to 15 are carriers of the disease allele.

Currently, imiglucerase (Cerezyme), miglustat (Zavesca), taliglucerase alfa (Elelyso), and velaglucerase (VPRIV) are the treatments available for Type 1 (non-neuropathic) Gaucher disease.

Alglucerase (Ceredase) is the first-generation enzyme replacement therapy (ERT) product for Gaucher's disease marketed by Genzyme Corp. (Cambridge, MA) and imiglucerase is Genzyme's second-generation Gaucher disease product. Alglucerase has been withdrawn from the market.

Cerezyme (imiglucerase) is an analogue of the human enzyme β -glucocerebrosidase, produced by recombinant DNA technology. β -Glucocerebrosidase is a lysosomal glycoprotein enzyme which catalyzes the hydrolysis of the glycolipid glucocerebroside to glucose and ceramide.

Cerezyme (imiglucerase) is indicated for long-term enzyme replacement in pediatric and adult members with a confirmed diagnosis of Type 1 Gaucher disease that results in one or more of the following conditions: anemia, thrombocytopenia, bone disease, hepatomegaly or splenomegaly.

They have not been shown to reverse or ameliorate neurological symptoms associated with Type 2 (acute neuronopathic) Gaucher disease. Alglucerase and imiglucerase have also not been shown to improve health outcomes in patients with Type 1 Gaucher disease without signs or symptoms of disease.

Enzyme replacement therapy also is used for individuals known to be at-risk for neuronopathic type 3 disease due to genotype or family history who present before the onset of neurologic symptoms. ERT may stabilize the neurologic progression in some of these patients or be effective only on the somatic signs and symptoms.

At this time, no effective minimum dosage has been established for alglucerase or imiglucerase. These drugs are administered no more than 3 times per week. The patient's response is reassessed at least every 3 months, with the intent of adjusting the frequency and size of doses.

Alglucerase and imiglucerase are administered by intravenous infusion over a period of 1 to 2 hours. Dosage should be individualized for each patient, based on disease severity and patient response. Cerezyme (imiglucerase) is supplied as a sterile, non-pyrogenic, lyophilized product and is available as 200 Units per vial and 400 Units per vial. According to the United States (U.S.) Food and Drug Administration (FDA) approved labeling for imiglucerase (Cerezyme), initial dosages range from 2.5 units per kilogram of body weight 3 times a week up to as much as 60 units/kg administered once two every weeks or as infrequently as every four weeks. Disease severity may dictate that treatment be initiated at a relatively high dose or relatively frequent administration. Dosage adjustments should be made on an individual basis and may increase or decrease, based on achievement of therapeutic goals as assessed by routine comprehensive evaluations of the patient's clinical manifestations. Doses as low as 1 unit/kg monthly may be adequate for some patients. The dose for which the most data are available is 60 units/kg every 2 weeks.

Disease severity may dictate that the drug be initiated with relatively high doses or relatively frequent administration. After patient response is well-established, a reduction in dosage may be attempted for maintenance therapy. Progressive reductions can be made at intervals of 3 to 6 months while carefully monitoring response parameters.

Doses up to 240 units/kg IV, administered once every two weeks have been used without reports of toxicity, according to the manufacturer. A specific maximum dosage is not available,

however, consensus guidelines state that doses > 60 units/kg IV once every two weeks are rarely necessary.

Although there are no known contraindications, Cerezyme (imiglucerase) should be used with caution in members with: a known hypersensitivity to imiglucerase; who are pregnant or breastfeeding; less than two years of age as data in this population is not available; or a high risk for pulmonary hypertension. Members with respiratory symptoms should be evaluated for the presence of pulmonary hypertension.

The use of Cerezyme (imiglucerase) should be directed by a qualified health care professional knowledgeable in the management of Gaucher's disease.

The International Collaborative Gaucher Group (ICGG) U.S. Regional Coordinators recommend that all children with Gaucher disease be treated with ERT. Children with Gaucher disease are at high-risk for irreversible, morbid complications. The diagnosis of Gaucher disease in the 1st and 2nd decades of life is indicative of a rapidly progressive course. Early intervention is necessary for these children, during the time when the skeleton is immature, to enable them to attain their peak skeletal mass by early adulthood.

Velaglucerase alfa (gene-activated human glucocerebrosidase) (Vpriv) for injection is an ERT for the treatment of Gaucher disease. Velaglucerase alfa is a hydrolytic lysosomal glucocerebrosidase-specific enzyme. Velaglucerase alfa catalyzes the hydrolysis of glucocerebroside, reducing the amount of accumulated glucocerebroside.

Velaglucerase is produced in a human cell line using gene-activation technology and has an identical amino acid sequence to the naturally occurring human enzyme. In contrast to imiglucerase, velaglucerase alfa contains the native human enzyme sequence. Kinetic parameters ($K(m)$ and $V(max)$) of velaglucerase alfa and

imiglucerase, as well as their specific activities, are similar. However, analysis of glycosylation patterns shows that velaglucerase alfa displays distinctly different structures from imiglucerase. The predominant glycan on velaglucerase alfa is a high-mannose type, with 9 mannose units, while imiglucerase contains a chitobiose tri-mannosyl core glycan with fucosylation. These differences in glycosylation affect cellular internalization; the rate of velaglucerase alfa internalization into human macrophages is at least 2-fold greater than that of imiglucerase (Brumshtein et al, 2010).

Shire Human Genetic Therapies, Inc. (2010) presented positive results from its first phase III study of velaglucerase alfa for the treatment of Type 1 Gaucher disease. This trial was a 12-month, randomized, double-blind, parallel-group global study in 25 treatment-naïve patients aged 2 years and older that evaluated velaglucerase alfa at dosages of 45 U/kg and 60 U/kg. The study's primary endpoint, mean hemoglobin concentration changes from baseline, was represented by a statistically significant increase of 23.3 % (+2.43 +/- 0.32 g/dL, $p < 0.0001$) at 12 months in the 60 U/kg group. Secondary endpoints for both doses were changes in platelet counts, changes in organ volumes, changes in surrogate markers of Gaucher disease, and for the 45 U/kg dose only, change in hemoglobin concentrations from baseline.

In January 2010, the FDA approved velaglucerase alfa for injection (VPRIV) to treat children and adults with Type I Gaucher disease. The approval was based on a priority review of data from 3 clinical studies of 82 patients aged 4 years and older, some of whom switched from imiglucerase therapy. The recommended velaglucerase regimen is 60 IU/kg administered every other week as a 1-hour intravenous infusion. The most common adverse reactions to VPRIV are allergic reactions. Other observed adverse reactions with VPRIV are headache, dizziness, abdominal pain, back pain, joint pain, nausea, fatigue/weakness, fever, and prolongation of activated partial thromboplastin time. Pediatric

patients were more likely than adults (greater than 10 % difference) to experience rash, upper respiratory tract infection, prolonged partial thromboplastin time, and pyrexia.

Treatment with Vpriv should be approached with caution in patients who have exhibited symptoms of hypersensitivity to the active ingredient or excipients in the drug product or to other enzyme replacement therapy.

Hypersensitivity reactions, including anaphylaxis, occurred in clinical studies and post-marketing experience. Hypersensitivity reactions were the most commonly observed adverse reactions in patients treated with VPRIV in clinical studies. Patients were not routinely pre-medicated prior to infusion of Vpriv during clinical studies. The most commonly observed symptoms of hypersensitivity reactions were: headache, dizziness, hypotension, hypertension, nausea, fatigue/asthenia, and pyrexia/body temperature increased. Generally the reactions were mild and, in treatment-naïve patients, onset occurred mostly during the first 6 months of treatment and tended to occur less frequently with time. Additional hypersensitivity reactions of chest discomfort, dyspnea, and pruritus have been reported in post-marketing experience.

As with any intravenous protein product, hypersensitivity reactions are possible, therefore appropriate medical support including personnel adequately trained in cardiopulmonary resuscitative measures and access to emergency measures should be readily available when VPRIV is administered. If anaphylactic or other acute reactions occur, immediately discontinue the infusion of VPRIV and initiate appropriate medical treatment. The management of hypersensitivity reactions should be based on the severity of the reaction, e.g., slowing the infusion rate, treatment with medications such as antihistamines, antipyretics and/or corticosteroids, and/or stopping and resuming treatment with increased infusion time. In cases where patients have exhibited symptoms of hypersensitivity to the active ingredient or excipients

in the drug product or to other enzyme replacement therapy, pre-treatment with antihistamines and/or corticosteroids may prevent subsequent reactions.

Velaglucerase alfa for injection is available as Vpriv 400 Units single-use vials as a sterile powder for reconstitution. The recommended dose is 60 Units/kg administered every other week as a 60-minute intravenous infusion. Members currently being treated with imiglucerase for type 1 Gaucher disease may be switched to Vpriv. Patients previously treated on a stable dose of imiglucerase are recommended to begin treatment with Vpriv at that same dose when they switch from imiglucerase to Vpriv. Dosage adjustments can be made based on achievement and maintenance of each member's therapeutic goals. Vpriv should be administered under the supervision of a healthcare professional.

Miglustat (Zavesca) is an oral ERT that has been approved by the FDA for the treatment of adult patients with mild-to-moderate Type 1 Gaucher disease for whom infusion/injection ERT is not a therapeutic option (e.g., due to constraints such as allergy, hypersensitivity, or poor venous access).

Zavesca (miglustat) is a competitive and reversible inhibitor of the enzyme glucosylceramide synthase, the initial enzyme responsible for the synthesis of glucosylceramide. Type 1 Gaucher patients are deficient in the enzyme glucocerebrosidase which is responsible for the degradation of glucosylceramide. Zavesca (miglustat) acts as a substrate reducer and allows the available glucocerebrosidase to act more efficiently.

In clinical studies, the most common adverse events due to miglustat included weight loss, diarrhea, and trembling in the hand (tremor). The most common serious adverse reaction was tingling or numbness in the hands or feet with or without pain (peripheral neuropathy). The labeling for Zavesca states that patients should undergo neurological examination at the start of

treatment and every 6 months thereafter, and that Zavesca should be re-assessed in patients who develop symptoms of peripheral neuropathy.

Approximately 85% of patients treated with Zavesca (miglustat) experienced diarrhea with or without weight loss. Incidence decreases over time and patients should avoid foods with high carbohydrate content. If diarrhea persists with usual interventions (e.g., diet modifications), the patient should be evaluated for underlying gastrointestinal disease.

Peripheral neuropathy has been reported and all patients should have neurological evaluations regularly.

Approximately 30% of patients treated with Zavesca (miglustat) reported tremors or exacerbation of existing tremor. Dose reduction may ameliorate tremors within days, but some patients will require discontinuation.

Pregnancy Category C: There are no adequate and well-controlled studies of Zavesca (miglustat) in pregnant women. Zavesca (miglustat) should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

Miglustat is available as Zavesca in 100 mg capsules. According to the labeling, the recommended dose for the treatment of adult patients with Type 1 Gaucher disease is one 100-mg capsule administered orally 3 times a day at regular intervals. Dosage may be reduced to 100 mg once or twice a day due to tremor or diarrhea.

Combination therapy with Cerezyme (imiglucerase) and Zavesca (miglustat) is not indicated.

Taliglucerase alfa (Elelyso) is a plant cell-derived recombinant human β -glucocerebrosidase for Gaucher disease.

Taliglucerase alfa is a hydrolytic lysosomal glucocerebroside-specific enzyme. Taliglucerase alpha catalyzes the hydrolysis of glucocerebroside to glucose and ceramide. Taliglucerase is thought to work by targeted delivery and uptake of tissue macrophages through the mannose lectin membrane system. Once inside the macrophage, taliglucerase degrades the accumulated glucocerebroside. Taliglucerase is not used to cure Gaucher disease, but to diminish the clinical manifestations of hepatosplenomegaly, and improve anemia and thrombocytopenia.

Elelyso (taliglucerase alfa) is indicated for long-term enzyme replacement therapy (ERT) for adults with a confirmed diagnosis of Type 1 Gaucher disease.

Zimran et al (2011) performed a phase III, double-blind, randomized, parallel-group, comparison-dose (30 versus 60 U/kg body weight/infusion) multi-national clinical trial. A 9-month, 20-infusion trial used inclusion/exclusion criteria in treatment-naïve adult patients with splenomegaly and thrombocytopenia. Safety end points were drug-related adverse events: Ab formation and hypersensitivity reactions. Primary efficacy end point was reduction in splenic volume measured by magnetic resonance imaging. Secondary end points were: changes in hemoglobin, hepatic volume, and platelet counts. Exploratory parameters included biomarkers and bone imaging. A total of 29 (11 centers) completed the protocol. There were no serious adverse events; drug-related adverse events were mild/moderate and transient. Two patients (6 %) developed non-neutralizing IgG Abs; 2 other patients (6 %) developed hypersensitivity reactions. Statistically significant spleen reduction was achieved at 9 months: 26.9 % (95 % confidence interval [CI]: -31.9 to -21.8) in the 30-unit dose group and 38.0 % (95 % CI: -43.4 to -32.8) in the 60-unit dose group (both $p < 0.0001$); and in all secondary efficacy end point measures, except platelet counts at the lower dose. These results support safety and effectiveness of taliglucerase alfa for Gaucher disease.

On May 1, 2012, the FDA approved taliglucerase alfa (Elelyso) for long-term enzyme replacement therapy to treat Type 1 Gaucher disease. According to the FDA, due to the small number of affected patients, the effectiveness of Elelyso was evaluated in a total of 56 patients with Type 1 Gaucher disease enrolled in 2 clinical trials. Many of these patients continued treatment on a longer-term extension study. In a multi-center, double-blind, parallel-dose trial, the effectiveness of Elelyso for use as an initial therapy was evaluated in 31 adult patients who had not previously received enzyme replacement therapy. Patients were randomly selected to receive Elelyso at a dose of either 30 or 60 units/kg. At both doses, Elelyso was effective in reducing spleen volume, the study's primary endpoint, from baseline by an average of 29 % in patients receiving the 30 units/kg dose and by an average of 40 % in patients receiving the 60 units/kg dose after 9 months of treatment. Improvements in liver volume, blood platelet counts, and red blood cell (hemoglobin) levels also were observed.

The effectiveness of Elelyso was evaluated in another study of 25 patients with Type 1 Gaucher disease who were switched from imiglucerase. In this multi-center, open-label, single-arm trial, patients who had been receiving treatment with imiglucerase for at least 2 years were switched to Elelyso infusions every other week at the same dose of imiglucerase. Results showed Elelyso was effective in maintaining spleen and liver volumes, blood platelet counts, and hemoglobin levels over a 9-month evaluation period.

The most common side effects associated with the use of Elelyso were infusion reactions and allergic reactions. Symptoms of infusion reactions include headache, chest pain or discomfort, weakness, fatigue, hives, skin redness, increased blood pressure, back pain, joint pain, and flushing. As with other intravenous protein products, anaphylaxis has been observed in some patients during Elelyso infusions. Other commonly observed side effects observed in greater than 10 % of patients treated with Elelyso

included arthralgia, back pain, extremity pain, headache, influenza, nasopharyngitis, upper respiratory tract infection, and urinary tract infections.

Anaphylaxis has been observed in some patients treated with Elelyso. If anaphylaxis occurs, immediately discontinue infusion and initiate appropriate treatment

The most commonly observed symptoms of infusion reactions (including allergic reactions) were headache, chest pain or discomfort, asthenia, fatigue, urticarial, erythema, increased blood pressure, back pain, arthralgia, and flushing. If allergic or infusion reactions occur, decreasing the infusion rate, temporarily stopping the infusion, or administering antihistamines and or antipyretics is recommended.

Taliglucerase alfa is available as Elelyso in 200 Unit single use vials for intravenous infusion. The recommended dose is 60 Units/kg of body weight administered every 2 weeks as a 60-120 minute intravenous infusion. Members currently being treated with imiglucerase for Type 1 Gaucher disease can be switched to Elelyso. Patients previously treated on a stable dose of imiglucerase are recommended to begin treatment with Elelyso at the same dose when they switch from imiglucerase to Elelyso.

Dose adjustments can be made based on achievement and maintenance of each member's therapeutic goals. Clinical studies have evaluated dose ranges from 11 Units/kg to 73 Units/kg every other week.

Cerdelga (eliglustat) is an inhibitor of glucosylceramide synthase, where glucosylceramide is the primary substrate of β -glucosidase, the enzyme lacking in Gaucher disease. Historically, Gaucher cells are characterized by engorged macrophages, as glucosylceramide

accumulates in these cells, causing cellular and organ dysfunction. Cerdelga (eliglustat) inhibits the formation of glucosylceramide, which helps to limit disease.

The U.S. Food and Drug Administration (FDA) approved eliglustat (Cerdelga) capsules for certain adult Gaucher disease type 1 patients (Genzyme, 2014) who are CYP2D6 poor metabolizers (PMs), intermediate metabolizers (IMs) or extensive metabolizers (EMs). Eliglustat is a specific ceramide analogue inhibitor of the enzyme glucosylceramide synthase (IC₅₀ = 10 ng/mL) with broad tissue distribution, which results in reduced production of glucosylceramide. Glucosylceramide is the substance that builds up in the cells and tissues of people with Gaucher disease.

The FDA approval was based on efficacy data from two positive Phase 3 studies for eliglustat: one in patients new to therapy (Trial 1), and the other in patients switching from approved enzyme replacement therapies (Trial 2) (Genzyme, 2014). The filing also incorporated four years of efficacy data from the Cerdelga Phase 2 study.

In Trial 1, improvements were seen across the following endpoints after 9 months on eliglustat: spleen size, platelet levels, hemoglobin levels, and liver volume (Genzyme, 2014). Patients continue to receive eliglustat in the extension period, and the majority of patients have been on treatment for over eighteen months. Trial 2 met the pre-specified criteria for non-inferiority to an enzyme replacement therapy (imiglucerase), which was a composite endpoint of each of the following parameters: spleen volume, hemoglobin levels, platelet counts, and liver volume. Patients continue to receive eliglustat in the extension period, and the majority of patients have been on treatment for over two years.

The most common adverse reactions (≥10%) are fatigue, headache, nausea, diarrhea, back pain, pain in extremities, and upper abdominal pain (Genzyme, 2014).

Eliglustat capsules are indicated for the long-term treatment of adults with Gaucher disease type 1 (GD1) who are CYP2D6 extensive metabolizers (EMs), intermediate metabolizers (IMs), or poor metabolizers (PMs) as detected by an FDA-cleared test (Genzyme, 2014). Patients who are CYP2D6 ultra-rapid metabolizers (URMs) may not achieve adequate concentrations of CERDELGA to achieve a therapeutic effect. The labeling of Cerdelga states that a specific dose cannot be recommended for those patients whose CYP2D6 genotype cannot be determined (indeterminate metabolizers).

Eliglustat is contraindicated in the following patients due to the risk of significantly increased eliglustat plasma concentrations which may result in prolongation of the PR, QTc, and/or QRS cardiac intervals that could result in cardiac arrhythmias: EMs or IMs taking a strong or moderate CYP2D6 inhibitor concomitantly with a strong or moderate CYP3A inhibitor and IMs or PMs taking a strong CYP3A inhibitor (Genzyme, 2014).

The labeling states that drugs that inhibit CYP2D6 and CYP3A may significantly increase the exposure to eliglustat; eliglustat dose adjustment may be needed, depending on metabolizer status (Genzyme, 2014).

Because eliglustat is predicted to cause increases in ECG intervals at substantially elevated plasma concentrations, use is not recommended in patients with pre-existing cardiac disease, long QT syndrome, or in combination with Class IA and Class III antiarrhythmic medications (Genzyme, 2014).

The labeling states that eliglustat should only be administered during pregnancy if the potential benefit justifies the potential risk; based on animal data, eliglustat may cause fetal harm (Genzyme, 2014). The labeling recommends discontinuing drug or nursing

based on importance of drug to mother. Eliglustat is not recommended in patients with moderate to severe renal impairment or in patients with hepatic impairment.

Cerdelga (eliglustat) is available as 84 mg hard gelatin capsules. The recommended dose of Cerdelga (eliglustat) is 84 mg twice daily in CYP 2D6 IMs and EMs, with or without food. In PMs, the recommended dose is 84 mg once daily. Cerdelga (eliglustat) should not be used in CYP2D6 ultra-rapid metabolizers as they may not receive adequate serum concentrations of drug.

Cerdelga (eliglustat) therapy should be avoided in persons with the following concomitant conditions:

- Concurrent use of a strong or moderate CYP2D6 inhibitor (eg. paroxetine, terbinafine) with a strong or moderate CYP3A inhibitor (eg. ketoconazole) in patients who are extensive metabolizers (EMs) or IMs (intermediate metabolizers)
- Concurrent use of a strong CYP3A inhibitor in patients who are intermediate metabolizers (IMs) or poor metabolizers (PMs) (eg. ketoconazole, fluconazole).

An UpToDate review on “Gaucher disease: Treatment” (Hughes, 2018) states that “Enzyme-enhancement therapy (EET), which attempts to increase the residual function of mutant enzymes, is another potential future therapy for GD. In EET, pharmacologic or chemical chaperones are used to stabilize folding of mutant glucocerebrosidase or decrease its degradation.

Chemical chaperones for the treatment of GD have been evaluated in clinical trials, but the development program of the lead compound GD (afegostat tartrate) was suspended as a result of lack of efficacy”.

Pastores and colleagues (2016) noted that anti-drug antibodies (ADA) may develop with biological therapies, possibly leading to a reduction of treatment efficacy and to allergic and other adverse reactions. Patients with Gaucher disease were tested for ADA every 6 or 12 weeks in clinical studies of velaglucerase alfa ERT, as part of a range of safety endpoints. In 10 studies between April 2004 and March 2015, a total of 289 patients aged 2 to 84 years (median of 43) were assessed for the development of anti-velaglucerase alfa antibodies; 64 patients were treatment-naïve at baseline and 225 patients were switched to velaglucerase alfa from imiglucerase treatment. They received velaglucerase alfa treatment for a median of 36.4 weeks (interquartile range [IQR] 26.4 to 155.4 weeks); 4 patients (1.4 %) became positive for anti-velaglucerase alfa IgG antibodies, 2 of whom had antibodies that were neutralizing in-vitro, but there were no apparent changes in patients' platelet counts, hemoglobin levels or levels of CCL18 and chitotriosidase, suggestive of clinical deterioration after anti-velaglucerase alfa antibodies were detected, and no infusion-related adverse events (AEs) were reported. The authors concluded that less than 2 % of patients exposed to velaglucerase alfa tested positive for antibodies and there was no apparent correlation between anti-velaglucerase alfa antibodies and AEs or pharmacodynamics or clinical responses.

Lysosomal Acid Lipase Deficiency

Kanuma (sebelipase alfa) is a recombinant human lysosomal acid lipase (rhLAL). It is produced by recombinant DNA technology in the egg white of eggs laid by genetically engineered chickens. The FDA approved sebelipase alfa (Kanuma) for the treatment of patients with a diagnosis of lysosomal acid lipase deficiency (LAL-D).

LAL-D is a genetic, chronic, and progressive metabolic disease that may be associated with significant morbidity and premature mortality. It is an ultra-rare disease, which is defined as a disease

that affects fewer than 20 patients/1,000,000 of the general population.

LAL-D is caused by genetic mutations that result in a marked decrease or loss in LAL enzyme activity in the lysosomes across multiple body tissues, leading to the chronic build-up of cholesteryl esters and triglycerides in the liver, blood vessel walls, and other organs. Without treatment, the youngest patients with LAL-D face rapid disease progression that is typically fatal within a matter of months. Infants experience profound growth failure, liver fibrosis, and cirrhosis, with a median age of death at 3.7 months. In an observational study, approximately 50 % of children and adults with LAL-D progressed to fibrosis, cirrhosis, or liver transplant in 3 years. The median age of onset of LAL-D is 5.8 years. Many older patients may be asymptomatic with increased risk of development of fibrosis, cirrhosis, liver failure, accelerated atherosclerosis, and cardiovascular disease.

There were no approved therapies available to treatment patients with LAL deficiency. The standard of care prior to approval of sebelipase alfa was supportive care, including lipid lowering therapies, vitamin E, hematopoietic stem cell (HSCT), and liver transplantation.

Measurement of LAL activity can be performed with a Dried Blood Spot (DBS) test using the fluorimetric substrate 4-methylumbelliferyl palmitate. Dried blood spot LAL enzyme testing was the primary methodology used in the clinical development program (including the pivotal trials).

Sebelipase alfa is an enzyme replacement therapy that addresses the underlying cause of lysosomal acid lipase deficiency (LAL-D) by reducing substrate accumulation in the lysosomes of cells throughout the body. Sebelipase alfa binds to cell surface receptors via glycans expressed on the protein and is subsequently internalized into lysosomes. Sebelipase alfa catalyzes the

lysosomal hydrolysis of cholesteryl esters and triglycerides to free cholesterol, glycerol and free fatty acids. In clinical studies, treatment with Kanuma improved survival in infants with LAL-D and led to significant reductions in ALT and liver fat content, as well as significant improvements in lipid parameters, in children and adults with LAL-D.

The FDA approval of sebelipase alfa was based on data from two clinical studies and a supporting open-label extension study comprising infant, pediatric, and adult patients with LAL-D. Study results showed significant benefit in terms of survival (67 %, or 6 out of 9) in patients with the infant form of LAL-D beyond 12 months, compared with 0 out of 21 patients in an untreated historical cohort. In pediatric and adult patients with LAL-D (aged 4 to 58 years), treatment with sebelipase alfa resulted in larger reductions from baseline in ALT values and liver fat content, as measured by MRI, compared to treatment with placebo. In addition, treated patients had significant improvements in lipid parameters, including LDL-C, HDL-C, non-HDL-C, and triglycerides, compared to placebo. The significance of these findings as they relate to cardiovascular morbidity and mortality or progression of liver disease in LAL deficiency has not been established.

The most commonly reported adverse events observed in clinical trials in infants were diarrhea, vomiting, fever, rhinitis, anemia, cough, nasopharyngitis, and urticaria. The most commonly reported adverse events observed in clinical trials in pediatric and adult patients were headache, fever, oropharyngeal pain, nasopharyngitis, asthenia, constipation, and nausea.

Hypersensitivity reactions, including anaphylaxis, have been reported in patients treated with sebelipase alfa. In clinical trials, 3 of 106 (3 %) patients treated with sebelipase alfa experienced signs and symptoms consistent with anaphylaxis. These patients experienced reactions during infusion with signs and symptoms

including chest discomfort, conjunctival injection, dyspnea, generalized and itchy rash, hyperemia, swelling of eyelids, rhinorrhea, severe respiratory distress, tachycardia, tachypnea, and urticaria. Anaphylaxis has occurred as early as the sixth infusion and as late as 1 year after treatment initiation.

In clinical trials, 21 of 106 (20 %) sebelipase-treated patients, including 9 of 14 (64 %) infants and 12 of 92 (13 %) pediatric patients, 4 years and older, and adults experienced signs and symptoms either consistent with or that may be related to a hypersensitivity reaction. Signs and symptoms of hypersensitivity reactions, occurring in 2 or more patients, included abdominal pain, agitation, fever, chills, diarrhea, eczema, edema, hypertension, irritability, laryngeal edema, nausea, pallor, pruritus, rash, and vomiting. The majority of reactions occurred during or within 4 hours of the completion of the infusion. Patients were not routinely pre-medicated prior to infusion of sebelipase in these clinical trials.

Administration of sebelipase alfa should be supervised by a healthcare provider knowledgeable in the management of hypersensitivity reactions including anaphylaxis. Moreover, the label advises providers to consider the risks and benefits of treatment of sebelipase alfa in patients with known systemic hypersensitivity reactions to eggs or egg products.

The most common adverse reactions listed in the Prescribing Information include the following:

- Infants with rapidly progressive LAL deficiency presenting within the first 6 months of life (30 % or more): diarrhea, vomiting, fever, rhinitis, anemia, cough, nasopharyngitis, and urticaria.
- Pediatric and adult patients with LAL deficiency (8 % or more): headache, fever, oropharyngeal pain, nasopharyngitis, asthenia, constipation, and nausea.

Mucopolysaccharidosis

Mucopolysaccharidosis refers to a group of inherited metabolic diseases which is caused by the absence or malfunctioning of certain enzymes the body needs to properly breakdown glycosaminoglycans, formerly called mucopolysaccharides, long chains of sugar molecules. As a result, sugars buildup in cells, blood and connective tissue which can lead to a variety of health problems. The mucopolysaccharidoses are classified within a larger group of disorders called lysosomal storage diseases. Seven distinct forms and numerous subtypes of mucopolysaccharidosis have been identified [MPS I (Hurler, Hurler-Scheie, Scheie), MPS II (Hunter syndrome), MPS III (Sanfilippo syndrome), MPS IV (Morquio syndrome), MPS VI (Maroteaux-Lamy syndrome), MPS VII (Sly syndrome), and MPS IX]. Associated signs and symptoms and the severity of the condition vary significantly by form. In general, most affected people appear healthy at birth and experience a period of normal development, followed by a decline in physical and/or mental function. As the condition progresses, it may affect appearance; physical abilities; organ and system functioning; and, in most cases, cognitive development. The underlying genetic cause varies by form. Most cases are inherited in an autosomal recessive manner, although one specific form (Type II) follows an X-linked pattern of inheritance. Treatment is based on the signs and symptoms present in each person (NIH, 2020).

Noh and Lee (2014) noted that MPSs are a group of rare inherited metabolic diseases caused by genetic defects in the production of lysosomal enzymes. Mucopolysaccharidoses are clinically heterogeneous and are characterized by progressive deterioration in visceral, skeletal and neurological functions. These investigators reviewed the classification and pathophysiology of MPSs and discussed current therapies and new targeted agents under development. They performed a Medline search through PubMed for relevant articles and treatment guidelines on MPSs published in

English for years 1970 to September of 2013 inclusive. The references listed in the identified articles, prescribing information of the drugs approved for the treatment of MPSs, as well as recent clinical trial information posted on Clinicaltrials.gov website, were reviewed. Until recently, supportive care was the only option available for the management of MPSs. In the early 2000s, ERT was approved by the FDA for the treatment of MPS I, II and VI.

Clinical trials of ERT showed substantial improvements in patients' somatic symptoms; however, no benefit was found in the neurological symptoms because the enzymes do not readily cross the blood-brain barrier (BBB). Hematopoietic stem cell transplantation (HSCT), another potentially curative treatment, is not routinely advocated in clinical practice due to its high risk profile and lack of evidence for efficacy, except in preserving cognition and prolonging survival in young patients with severe MPS I. In recent years, substrate reduction therapy (SRT) and gene therapy have been rapidly gaining greater recognition as potential therapeutic avenues. Substrate reduction therapy uses an orally available, small molecule drug (e.g., miglustat or eliglustat) that inhibits the first committed step in glycosphingolipid biosynthesis. The objective is to reduce the rate of biosynthesis of glycosphingolipids to offset the catabolic defect, restoring the balance between the rate of biosynthesis and the rate of catabolism. The authors concluded that ERT is effective for the treatment of many somatic symptoms, particularly walking ability and respiratory function, and remains the mainstay of MPS treatment. They stated that the usefulness of HSCT has not been established adequately for most MPSs. Although still under investigation, SRT and gene therapy are promising MPS treatments that may prevent the neurodegeneration not affected by ERT.

Parenti et al (2014) stated that pharmacological chaperone therapy is an emerging approach to treat lysosomal storage diseases.

Small-molecule chaperones interact with mutant enzymes, favor their correct conformation and enhance their stability. This approach showed significant advantages when compared with

existing therapies, particularly in terms of the bioavailability of drugs, oral administration and positive impact on the quality of patients' lives. On the other hand, future research in this field must confront important challenges. The identification of novel chaperones is indispensable to expanding the number of patients amenable to this treatment and to optimize therapeutic efficacy. It is important to develop new allosteric drugs, to address the risk of inhibiting target enzymes. The authors concluded that future research must also be directed towards the exploitation of synergies between chaperone treatment and other therapeutic approaches.

Parenti et al (2015) stated that lysosomal storage diseases are a group of rare, inborn, metabolic errors characterized by deficiencies in normal lysosomal function and by intra-lysosomal accumulation of un-degraded substrates. The past 25 years have been characterized by remarkable progress in the treatment of these diseases and by the development of multiple therapeutic approaches. These approaches include strategies aimed at increasing the residual activity of a missing enzyme (ERT, HSCT, pharmacological chaperone therapy and gene therapy) and approaches based on reducing the flux of substrates to lysosomes.

Giugliani et al (2016) stated that mucopolysaccharidoses MPSs are progressive, usually severe, and, in a significant number of cases, involve cognitive impairment. These investigators noted that this review would not cover established treatments (e.g., bone marrow/hematopoietic stem cell transplantation and classic intravenous ERT), whose long-term outcomes have already been published (MPS I, MPS II, and MPS VI), instead, it focused on emerging therapies for MPSs, which includes intravenous ERT for MPS IVA and VII, intrathecal ERT, ERT with fusion proteins, substrate reduction therapy, gene therapy, and other novel approaches.

Mucopolysaccharidosis I (Hurler, Hurler-Scheie, and Scheie)

The mucopolysaccharidoses (MPSs) are a group of inherited lysosomal storage disorders caused by the deficiency of specific enzymes that are required for the degradation of glycosaminoglycans (GAGs), or mucopolysaccharides. Mucopolysaccharidosis I disease (MPS I) is a progressive, autosomal recessive genetic disorder resulting from a defect in the gene for the lysosomal enzyme alpha-L-iduronidase. It is estimated that approximately 1,000 persons are afflicted with MPS I in the U.S. This enzyme deficiency results in an accumulation of the glycosaminoglycans dermatan sulfate and heparan sulfate, which are components of the extracellular matrix and connective tissues throughout the body. The inability to catabolize GAG results in its accumulation in the lysosome, resulting in cell, tissue, and organ dysfunction. Mucopolysaccharidosis I disease results in a variety of clinical manifestations, including umbilical and inguinal hernias, skeletal abnormalities, recurrent and persistent upper respiratory tract infections, coarse facial features, arthropathy, hydrocephalus, spinal root and peripheral nerve entrapment, obstructive airway disease, sleep apnea, hearing loss, hepatosplenomegaly, corneal clouding, glaucoma, retinal degeneration, optic atrophy, cardiac valvular and ischemic myocardial damage. The diagnosis of MPS I is established by enzyme assay that measures alpha-L-iduronidase activity in leukocytes, plasma, or cultured skin fibroblasts. Enzyme activity is markedly deficient (less than 1 % normal) in affected patients. Prenatal diagnosis by measurement of alpha-L-iduronidase activity in cultured amniocytes or chorionic villi is also possible.

MPS I has been divided into three broad groups based on severity of symptoms - Hurler, Hurler-Scheie, and Scheie. Scheie syndrome was previously known as MPS V before being included in MPS I. MPS I may be viewed as a continuous spectrum of disease, with the most severely affected individuals on one end (Hurler syndrome), the less severely affected (attenuated) on the other end

(Scheie syndrome), and a wide range of different severities in between. All individuals with MPS I have an absence of, or insufficient levels of, the enzyme alpha-L-iduronidase, which is needed to break down glycosaminoglycans (NIH, 2020).

Aldurazyme (laronidase) is an enzyme replacement therapy for the treatment of mucopolysaccharidosis I (MPS I), a rare, autosomal recessive lysosomal storage disorder caused by a deficiency of the enzyme alpha-L-iduronidase. The lack of the enzyme causes glycosaminoglycans (GAG) to build up in cells. Manifestations of the disorder can include growth and developmental delay, enlargement of spleen and liver, skeletal deformity, cardiac and pulmonary impairment, vision or hearing loss and mental dysfunction.

Aldurazyme (laronidase) is indicated for enzyme replacement in patients with the Hurler and Hurler-Scheie forms of MPS I, and for Scheie patients with moderate to severe symptoms.

Laronidase is administered to provide exogenous enzyme for uptake into the lysosomes in order to increase the catabolism of GAG. Enzyme replacement therapy with laronidase has been shown to provide clinically important benefits, such as improved pulmonary function and walking ability and reduction of excess carbohydrates stored in organs. In a randomized, placebo-controlled clinical study for the FDA's approval of laronidase, 45 MPS I patients were randomly assigned to laronidase or placebo. After 26 weeks, patients treated with laronidase showed statistically significant improvement in forced vital capacity (FVC) (median difference of 2 % (95 % CI: 0.4 to 7)) compared to placebo-treated patients. In addition, laronidase-treated patients showed a trend toward improvement in distance walked in 6 mins (median difference 39 meters (95 % CI: -2 to 79) compared to placebo-treated patients; however, this difference was not statistically significant. Liver size and urinary GAG levels decreased in patients treated with laronidase compared to patients treated with placebo.

All 45 patients received open-label laronidase for 36 weeks following the double-blind period. Maintenance of mean FVC and an additional increase in mean distance walked in 6 mins were observed compared to the start of the open-label period among patients who were initially randomized to and then continued to receive laronidase. Among patients who had been initially randomized to placebo, improvements from baseline in mean FVC and distance walked in 6 mins were observed compared to the start of the open-label period.

Aldurazyme (laronidase) is available as 0.58 mg/mL solution for injection in a 5mL vial. Laronidase is administered intravenously once-weekly. The recommended dosage for laronidase is 0.58 mg/kg of body weight administered once weekly as an intravenous infusion. Pretreatment with antipyretics and/or antihistamines is recommended 60 minutes prior to the start of the infusion. There is no published information on the effect on response rates of increased doses of laronidase beyond the FDA-recommended dosage. As a prerequisite for approval, the FDA has required the manufacturer to conduct post-marketing studies of different dosages and schedules of laronidase in clinical response. At this point, there is no evidence to support dosing laronidase beyond that recommended in the product labeling.

The following are the most common adverse reactions (10% or more) associated with laronidase:

- 6 months of age and older include: infusion reactions (pyrexia, chills, blood pressure increased, tachycardia, and oxygen saturation decreased)
- 6 years and older include rash, upper respiratory tract infection, injection site reaction, hyperreflexia, paresthesia, flushing, and poor venous access.

Aldurazyme (laronidase) should be used with extreme caution and close monitoring, if at all, in any patient who has exhibited prior laronidase hypersensitivity.

Corrective surgery may be necessary for MPS I patients with joint contractures or foot and hand deformities. Corneal transplants may be required if vision problems become severe.

Grewal and colleagues (2005) described their initial experience with combined use of laronidase and hematopoietic stem cell transplantation in the treatment of patients with Hurler syndrome (n = 12). They concluded that in children with Hurler syndrome, laronidase and hematopoietic stem cell transplantation is feasible and well-tolerated. Development of antibodies against exogenous enzyme does not appear to correlate with infusion reactions or response to laronidase. A prospective study is needed to ascertain the effect of concomitant ERT on transplant outcomes.

Xue and colleagues (2016) stated that ERT with laronidase has an important role in the treatment of patients with MPS I. Laronidase is safe and has demonstrated effectiveness in terms of stabilizing or improving conventional clinical and laboratory markers of the disease. However, like most ERTs, laronidase produces an anti-drug IgG antibody response in more than 90 % of patients during the first few months of treatment. Pre-clinical data from the MPS I canine model suggested that ADA impair enzyme uptake in target tissues. In patients, the effects on tissue glycosaminoglycan (GAG) clearance are difficult to evaluate directly; but data from clinical studies have suggested an association between ADA and both a reduced pharmacodynamics response and hypersensitivity reactions. The authors carried out a comprehensive meta-analysis of pooled data from patients in 3 clinical studies of laronidase (including 1 study with an extension) to provide a more robust assessment of the relationship between the ADA response to laronidase, clinical and laboratory markers of MPS I, and hypersensitivity reactions. The meta-analysis demonstrated an

inverse relationship between the ADA response and the percent reduction in urinary GAG (uGAG) levels. However, no relationships between the ADA response and changes in percent predicted forced vital capacity and 6-minute walk test were seen. The study also re-assayed stored serum samples from the original trials with a novel method to determine the inhibitory effect of ADA. Patients with higher ADA exposure over time were found to have higher inhibition of enzyme uptake into cells. The authors concluded that high ADA exposure could result in a commensurate level of enzyme uptake inhibition that decreases the pharmacodynamics effect of the exogenously administered therapeutic enzyme, but with no clear effect on clinical efficacy.

Mucopolysaccharidosis II (Hunter Syndrome)

Mucopolysaccharidosis II (also called Hunter syndrome) is an X-linked, recessive, lysosomal storage disease that is caused by a defect of the iduronate-2-sulfatase gene, which breaks down the glycosaminoglycans heparin sulfate and dermatan sulfate inside cells. MPS II is the only mucopolysaccharidosis disorder in which the mother alone can pass the defective gene to a son (called X-linked recessive) (NIH, 2020). Thus, females with this condition are rare. It is diagnosed in approximately 1 out of 65,000 to 132,000 births. Hunter syndrome usually becomes apparent in children 1 to 3 years of age. Symptoms include growth delay, joint stiffness, and coarsening of facial features. In severe cases, patients experience neurological deficits, enlargement of the liver and spleen, cardiac as well as respiratory problems, and death.

On July 24, 2006, the FDA approved idursulfase (Elaprase) (Shire Human Genetic Therapies, Inc., Cambridge, MA) for the treatment of Hunter syndrome. Idursulfase was designated as an orphan drug, and was approved after a randomized, double-blind, placebo-controlled clinical trial of 96 patients showed that the treated subjects had an improved capacity to walk. At the end of the 53-week study, patients who received idursulfase infusions

experienced on average a 38-yard greater increase in the distance walked in 6 mins compared to the patients on placebo. The most serious side effects reported during the trial were hypersensitivity reactions to idursulfase that could be life-threatening. They included respiratory distress, drop in blood pressure, and seizure. Other frequent, but less serious side effects included fever, headache, and joint pain. The recommended dosage regimen of idursulfase is 0.5 mg/kg administered every week as an intravenous infusion.

Muenzer et al (2012) stated that intravenous ERT with idursulfase for Hunter syndrome has not been demonstrated to and is not predicted to cross the blood-brain barrier. Nearly all published experience with ERT with idursulfase has therefore been in patients without cognitive impairment (attenuated phenotype). Little formal guidance is available on the issues surrounding ERT in cognitively impaired patients with the severe phenotype. An expert panel was therefore convened to provide guidance on these issues. The clinical experience of the panel with 66 patients suggested that somatic improvements (e.g., reduction in liver volume, increased mobility, and reduction in frequency of respiratory infections) may occur in most severe patients. Cognitive benefits have not been seen. It was agreed that, in general, severe patients are candidates for at least a 6- to 12-month trial of ERT, excluding patients who are severely neurologically impaired, those in a vegetative state, or those who have a condition that may lead to near-term death. It is imperative that the treating physician discuss the goals of treatment, methods of assessment of response, and criteria for discontinuation of treatment with the family before ERT is initiated. The authors concluded that the decision to initiate ERT in severe Hunter syndrome should be made by the physician and parents; and must be based on realistic expectations of benefits and risks, with the understanding that ERT may be withdrawn in the absence of demonstrable benefits.

Muenzer et al (2016) noted that approximately 2/3 of patients with the lysosomal storage disease MPS II have progressive cognitive impairment. Intravenous (i.v.) ERT does not affect cognitive impairment because recombinant iduronate-2-sulfatase (idursulfase) does not penetrate the BBB at therapeutic concentrations. In a phase I/II study, these researchers examined the safety of idursulfase formulated for intrathecal administration (idursulfase-IT) via intrathecal drug delivery device (IDDD). A secondary end-point was change in concentration of glycosaminoglycans in cerebro-spinal fluid (CSF). A total of 16 cognitively impaired males with mucopolysaccharidosis II who were previously treated with weekly i.v. idursulfase 0.5 mg/kg for greater than or equal to 6 months were enrolled. Patients were randomized to no treatment or 10-mg, 30-mg, or 1-mg idursulfase-IT monthly for 6 months (4 patients per group) while continuing i.v. idursulfase weekly. No serious adverse events related to idursulfase-IT were observed. Surgical revision/removal of the IDDD was required in 6 of 12 patients. Twelve total doses were administered by lumbar puncture. Mean CSF glycosaminoglycan concentration was reduced by approximately 90 % in the 10-mg and 30-mg groups and approximately 80 % in the 1-mg group after 6 months. The authors concluded that these preliminary data supported further development of investigational idursulfase-IT in MPS II patients with the severe phenotype who have progressed only to a mild-to-moderate level of cognitive impairment.

In a 109-week, non-randomized, observational study of MPS II patients already enrolled in the Hunter Outcome Survey (HOS), Giugliani and associates (2017) evaluated the long-term immunogenicity of idursulfase, and examined the effect of idursulfase-specific antibody generation on treatment safety (via infusion-related adverse events [IRAEs]) and pharmacodynamics (via urinary glycosaminoglycans [uGAGs]). Male patients greater than or equal to 5 years, enrolled in HOS regardless of idursulfase treatment status were eligible. Blood/urine samples for anti-idursulfase antibody testing and uGAG measurement were

collected every 12 weeks. Due to difficulties in enrolling treatment-naïve patients, data collection was limited to 26 enrolled patients of 100 planned patients (aged 5.1 to 35.5 years) all of whom were non-naïve to treatment; 15 (58 %) patients completed the study.

There were 11/26 (42 %) sero-positive patients at baseline (Ab +), and 2/26 (8 %) others developed intermittent sero-positivity by week 13. A total of 9/26 patients (35 %) had greater than or equal to 1 sample positive for neutralizing antibodies. Baseline uGAG levels were low due to prior idursulfase treatment and did not change appreciably thereafter. Ab + patients had persistently higher uGAG levels at entry and throughout the study than Ab – patients; 9 of 26 (34 %) patients reported IRAEs. Ab + patients appeared to have a higher risk of developing IRAEs than Ab – patients. However, the relative risk (RR) was not statistically significant and decreased after adjustment for age. The authors concluded that 50 % of study patients developed idursulfase antibodies; notably Ab + patients had persistently higher average uGAG levels; however, a clear association between IRAEs and antibodies was not established.

Mucopolysaccharidosis IV (Morquio Syndrome)

Mucopolysaccharidosis IV (MPS IV A and B), also known as Morquio syndrome, is an autosomal recessive mucopolysaccharide storage disease. There are 2 forms of Morquio syndrome, Type A and Type B, with similar clinical findings and autosomal inheritance. MPS IV A results from mutations in the gene encoding galactosamine-6-sulfatase (GALNS), located at 16q24.3. MPS IV B is due to beta-galactosidase deficiency. The clinical features result from accumulation of keratan sulfate and chondroitin-6-sulfate.

The enzymes deficient in Morquio syndrome (mucopolysaccharidosis type IV) are galactosamine-6-sulfatase (ie, N -acetyl-galactosamine-6-sulfate sulfatase) and β -galactosidase.

The diagnosis is confirmed by direct enzymatic assay in leukocytes or fibroblasts.

Deficiency of the enzyme results in excessive lysosomal storage of keratan sulfate in many tissues and organs. This accumulation causes systemic skeletal dysplasia, short stature, and joint abnormalities, which limit mobility and endurance. Malformation of the thorax impairs respiratory function, and malformation of neck vertebrae and ligament weakness causes cervical spinal instability and, potentially, cord compression. Other symptoms may include hearing loss, corneal clouding, and heart valve disease. Morquio A syndrome is estimated to occur in 1 in 200,000 to 300,000 live births. Symptoms usually start between ages 1 and 3.

Vimizim (elosulfase alfa) is a hydrolytic lysosomal glycosaminoglycan (GAG)-specific enzyme. The FDA has approved elosulfase alfa (Vimizim) for persons with Mucopolysaccharidosis type IVA (MPS IVA; Morquio A syndrome). The FDA approval of elosulfase was based on a randomized controlled clinical trial and an uncontrolled extension study.

A 24-week, randomized, double-blind, placebo-controlled clinical trial of elosulfase alfa was conducted in 176 patients with MPS IVA, aged 5 to 57 years. Patients were randomized to three treatment groups: elosulfase alfa 2 mg/kg once-weekly (n = 58), elosulfase alfa 2 mg/kg once every other week (n = 59), or placebo (n = 59). All patients were treated with antihistamines prior to each infusion. The primary end-point was the change from baseline in the distance walked in 6 minutes (6 minute walk test, 6-MWT) at Week 24. The other endpoints included changes from baseline in the rate of stair climbing in 3 minutes (3-minute stair climb test, 3-MSCT) and changes from baseline in urine keratan sulfate (KS) levels at Week 24. The treatment effect in the distance walked in 6 minutes, compared to placebo, was 22.5 m (95 % CI: 4.0 to 40.9; p = 0.0174) in patients who received elosulfase 2 mg/kg once-weekly. There was no difference in the rate of stair climbing between

patients who received elosulfase 2 mg/kg once-weekly and those who received placebo. Patients who received elosulfase 2 mg/kg once every other week performed similarly in the 6-MWT and 3-MSCT as those who received placebo. The reduction in urinary KS levels from baseline, a measure of pharmacodynamic effect, was greater in the elosulfase treatment groups compared to placebo. The FDA labeling states that the relationship between urinary KS and other measures of clinical response has not been established.

Patients who participated in the placebo-controlled trial were eligible to continue treatment in an open-label extension trial; 173 of 176 patients enrolled in the extension trial in which patients received elosulfase 2 mg/kg once-weekly (n = 86) or elosulfase 2 mg/kg once every other week (n = 87). In patients who continued to receive elosulfase 2 mg/kg once-weekly for another 48 weeks (for a total of 72-week exposure), walking ability showed no further improvement beyond the first 24 weeks of treatment in the placebo-controlled trial.

The most common side effects in patients treated with elosulfase alfa during clinical trials included fever, vomiting, headache, nausea, abdominal pain, chills and fatigue. The FDA approved labeling states that the safety and effectiveness of elosulfase alfa have not been established in pediatric patients less than 5 years of age.

Elosulfase alfa was approved with a boxed warning to include the risk of anaphylaxis. During clinical trials, life-threatening anaphylactic reactions occurred in some patients during Vimizim infusions. Anaphylaxis, presenting as cough, erythema, throat tightness, urticaria, flushing, cyanosis, hypotension, rash, dyspnea, chest discomfort, and gastrointestinal symptoms in conjunction with urticaria, have been reported to occur during infusions, regardless of duration of the course of treatment. Closely observe patients during and after Vimizim administration and be prepared to manage anaphylaxis. Inform patients of the signs and symptoms of

anaphylaxis and have them seek immediate medical care should symptoms occur. Patients with acute respiratory illness may be at risk of serious acute exacerbation of their respiratory compromise due to hypersensitivity reactions, and require additional monitoring.

Elosulfase alfa is available as Vimizim in 5 mg/5 mL single-use vials for intravenous infusion. The recommended dose of elosulfase alfa is 2 mg/kg given intravenously over a minimum range of 3.5 to 4.5 hours, based on infusion volume, once every week.

Melton and colleagues (2017) stated that many ERTs for lysosomal storage disorders use the cell-surface cation-independent mannose-6 phosphate receptor (CI-M6PR) to deliver ERTs to the lysosome. However, neutralizing antibodies (NAb) may interfere with this process. These researchers previously reported that most individuals with Morquio A who received elosulfase alfa in the phase III MOR-004 trial tested positive for NABs capable of interfering with binding to CI-M6PR ectodomain in an ELISA-based assay. However, no correlation was detected between NAb occurrence and clinical efficacy or pharmacodynamics. To quantify and better characterize the impact of NABs, these investigators developed a functional cell-based flow cytometry assay with a titer step that detects antibodies capable of interfering with elosulfase alfa uptake. Serum samples collected during the MOR-004 trial were tested and titers were determined. Consistent with earlier findings on NAb positivity, no correlations were observed between NAb titers and the clinical outcomes of elosulfase alfa-treated individuals with Morquio A.

Mucopolysaccharidosis VI (Maroteaux-Lamy Syndrome)

Naglazyme (galsulfase) is a hydrolytic lysosomal glycosaminoglycan (GAG)-specific enzyme indicated for patients with Mucopolysaccharidosis VI (MPS VI; Maroteaux-Lamy syndrome). MPS VI (Maroteaux-Lamy syndrome) is an autosomal recessive disorder that results from the deficiency arylsulfatase B

activity which is necessary for the breakdown of the glycosaminoglycans (GAGs). Cellular accumulation of GAG residues disrupt normal cell function, and result in progressive cellular, tissue, and organ dysfunction. Naglazyme (galsulfase) is indicated for the treatment of mucopolysaccharidosis VI (MPS VI; Maroteaux-Lamy syndrome).

Mucopolysaccharidosis VI (MPS VI), also known as Maroteaux-Lamy syndrome, is a debilitating, life-threatening genetic disease caused by a deficiency of the enzyme N-acetylgalactosamine 4-sulfatase. This deficiency results in the accumulation of glycosaminoglycans in the lysosomes, giving rise to progressive cellular, tissue and organ system dysfunction. An estimated 1,100 persons in the developed world have MPS VI. While some patients have rapidly progressing disease, others do not present with signs and symptoms until adolescence. Yet, MPS VI-related multisystemic abnormalities and significant functional disability are life-altering. The majority of patients with MPS VI die from disease-related complications between childhood and early adulthood.

Harmatz and associates (2004) assessed the safety and effectiveness of weekly treatment with human recombinant N-acetylgalactosamine 4-sulfatase (rhASB) in humans with MPS VI. Patients were randomized to weekly infusions of either high (1.0 mg/kg) or low (0.2 mg/kg) doses of rhASB. Six patients (3 males, 3 females; aged 7 to 16 years) completed at least 24 weeks of treatment, 5 of this group have completed at least 48 weeks. No drug-related serious side effects, significant laboratory abnormalities, or allergic reactions were observed in the study. The high-dose group experienced a more rapid and larger relative reduction in urinary glycosaminoglycan that was sustained through week 48. Improvements in the 6-min walk test were observed in all patients with dramatic gains in those walking less than 100 meters at baseline. Shoulder range of motion improved in all patients at week 48 and joint pain improved in patients with significant pain at baseline. The authors concluded that rhASB treatment was well-

tolerated and reduced lysosomal storage as indexed by a dose-dependent reduction in urinary glycosaminoglycan. Clinical responses were present in all patients, but the largest gains occurred in patients with advanced disease receiving high-dose rhASB.

On June 1, 2005, galsulfase (Naglazyme, BioMarin Pharmaceutical Inc., Novato, CA) was granted orphan drug status by the FDA for the treatment of MPS VI. Galsulfase has been reported to improve endurance as shown by the 12-min walk test as well as the 3-min stair climb. It reduces the urinary excretion of glycosaminoglycans, an indication of enzymatic bioactivity, in patients with MPS VI.

Naglazyme (galsulfase) is available in a 1mg/mL solution for injection in a 5mL vial. The recommended dose of Naglazyme (galsulfase) is 1 mg per kg of body weight administered once weekly as an intravenous infusion.

The most common adverse reactions (10% or more) include rash, pain, urticaria, pyrexia, pruritus, chills, headache, nausea, vomiting, abdominal pain, and dyspnea. The most common adverse reactions requiring interventions are infusion-related reactions.

Mucopolysaccharidoses VII (Sly Syndrome)

MPS VII is an autosomal recessive lysosomal storage disorder caused by deficiency of beta-glucuronidase (NORD, 2017). The deficiency in beta-glucuronidase leads to the accumulation of mucopolysaccharides in many tissues and organs of the body. MPS VII affects less than 150 patients worldwide (FDA, 2017). The features of MPS VII vary, but most affected individuals have skeletal abnormalities that become more pronounced with age, including short stature and changes in bones visible on X-rays (dysostosis multiplex). Patients can also develop aortic regurgitation, hepatomegaly, splenomegaly, and narrowed airways which can lead to pulmonary infections and trouble breathing. The

life expectancy of individuals with MPS VII depends on the severity of symptoms. Some affected individuals do not survive infancy, while others may live into adolescence or adulthood. Heart disease and airway obstruction are major causes of death in people with MPS VII. Affected individuals may have developmental delay and progressive intellectual disability.

Urinary levels of the mucopolysaccharides (dermatan sulfate, heparan sulfate, and chondroitin sulfate) are increased in affected individuals (NORD, 2017). The diagnosis of MPS VII may be confirmed by specialized tests that measure the level of beta-glucuronidase activity in blood or skin cells. Molecular genetic testing for mutations in the GUSB gene is available to confirm the diagnosis.

The U.S. Food and Drug Administration (FDA) approved vestronidase alfa-vjvk (Mepsevii) to treat pediatric and adult patients with an MPS VII (FDA, 2017). The safety and efficacy of vestronidase alfa-vjvk were established in clinical trial and expanded access protocols enrolling a total of 23 patients ranging from 5 months to 25 years of age. Patients received treatment with vestronidase alfa-vjvk at doses up to 4 mg/kg once every two weeks for up to 164 weeks. Efficacy was primarily assessed via the six-minute walk test in ten patients who could perform the test. After 24 weeks of treatment, the mean difference in distance walked relative to placebo was 18 meters. Additional follow-up for up to 120 weeks suggested continued improvement in three patients and stabilization in the others. Two patients in the vestronidase alfa-vjvk development program experienced marked improvement in pulmonary function. The product labeling for Mepsevii states that the effect of vestronidase alfa-vjvk on the central nervous system manifestations of MPS VII has not been determined.

The FDA is requiring the manufacturer to conduct a post-marketing study to evaluate the long-term safety of the product (FDA, 2017).

The most common side effects after treatment with vestronidase alfa-vjvk include infusion site reactions, diarrhea, rash and anaphylaxis (FDA, 2017).

The labeling includes a black box warning about anaphylaxis with Mepsevii administration (Ultragenyx, 2017).

Mepsevii is available as 10 mg/5 mL (2 mg/mL) in a single-dose vial (Ultragenyx, 2017). The recommended dosage is 4 mg/kg administered every two weeks as an intravenous infusion.

Neuronal Ceroid Lipofuscinosis type 2 (CLN2)

The U.S. Food and Drug Administration (FDA) approved cerliponase alfa (Brineura) to slow the loss of ambulation in symptomatic pediatric patients 3 years of age and older with late infantile neuronal ceroid lipofuscinosis type 2 (CLN2), also known as tripeptidyl peptidase 1 (TPP1) deficiency, a form of Batten disease.

CLN2 disease is an rare, rapidly progressive fatal brain condition, which affects less than one in one million U.S. residents. Every year approximately 20 children are born in the U.S. with CLN2 disease. These affected children completely lose the ability to walk and talk around 6 years of age. During the later stages of the disease, feeding and tending to everyday needs become very difficult with death often occurring between 8 and 12 years of age.

Children with CLN2 disease typically begin experiencing seizures between the ages of 2 and 4 years old, preceded in the majority of cases by language development delay. The disease progresses rapidly with most affected children losing the ability to walk and talk by approximately 6 years of age. Initial symptoms are followed by movement disorders, motor deterioration, dementia, blindness, and

death usually occurring between the ages of 8 and 12 years of age. During the later stages of the disease, feeding and tending to everyday needs become very difficult.

The neuronal ceroid lipofuscinoses (NCLs) are a heterogeneous group of lysosomal storage disorders that includes the autosomal recessive neurodegenerative disorder CLN2 disease. CLN2 disease is caused by mutations in the TPP1 gene resulting in deficient activity of the enzyme tripeptidyl peptidase 1 (TPP1). In the absence of TPP1, lysosomal storage materials normally metabolized by this enzyme accumulate in many organs, particularly in the brain and retina. Buildup of these storage materials in the cells of the nervous system contribute to the progressive and relentless neurodegeneration which manifests as loss of cognitive, motor, and visual functions.

Cerliponase alfa is a recombinant form of human tripeptidyl peptidase 1 (TPP1), the enzyme deficient in patients with CLN2 disease. It is an enzyme replacement therapy designed to restore TPP1 enzyme activity and break down the storage materials that cause CLN2 disease. In order to reach the cells of the brain and central nervous system, the treatment is delivered directly into the fluid surrounding the brain (cerebrospinal fluid) using BioMarin's patented technology.

Cerliponase alfa is a prescription medication used to slow loss of ability to walk or crawl (ambulation) in symptomatic pediatric patients 3 years of age and older with late infantile neuronal ceroid lipofuscinosis type 2 (CLN2), also known as tripeptidyl peptidase 1 (TPP1) deficiency. Cerliponase alfa is only given by infusion into the fluid of the brain (known as an intraventricular injection) and using sterile technique to reduce the risk of infection. An intraventricular access device or port must be in place at least 5 to 7 days prior to the first infusion. Intraventricular access device-

related infections were observed with cerliponase alfa treatment. The child is to be monitored for any signs of infection. The child's intraventricular access device may need to be replaced over time.

In clinical trials, cerliponase alfa, an enzyme replacement therapy, was shown to slow the loss of ambulation in symptomatic pediatric patients 3 years of age and older with CLN2 disease. It is the first enzyme replacement therapy to be directly administered to the brain, treating the underlying cause of the condition by replacing the deficient TPP1 enzyme. Using an established technique most often used in oncology – intraventricular administration – the therapy is delivered directly into fluid surrounding the brain, known as the cerebrospinal fluid.

The approval was supported by safety and efficacy data assessed over 96 weeks in a non-randomized, single-arm dose escalation clinical study of patients with CLN2 disease. Cerliponase alfa treated patients were compared to untreated patients from a natural history cohort.

Patients were assessed for decline in the motor domain of the CLN2 Clinical Rating Scale. The scale measures performance of mobility with normal function being a score of 3 and no function being a score of 0. Decline was defined as having a sustained 2-point decline or an unreversed score of 0 in the motor domain of the CLN2 Clinical Rating Scale.

Twenty-four patients aged 3-8 years were enrolled in the clinical study. One patient withdrew after week 1 due to inability to continue with study procedures; 23 patients were treated with cerliponase alfa every other week for 48 weeks and continued treatment during the extension.

Results from the 96-week analysis demonstrated the odds of cerliponase alfa -treated patients not having a decline were 13 times the odds of natural history cohort patients not having a

decline (Odds Ratio (95% Confidence Interval): 13.1 (1.2, 146.9)).

Of the 22 patients treated with cerliponase alfa and evaluated for efficacy at week 96, 21 (95%) did not decline, and only the patient who terminated early was deemed to have a decline in the motor domain of the CLN2 Clinical Rating Scale. In comparison, 50% of patients in an independent natural history cohort demonstrated progressive decline in motor function. Two cerliponase alfa treated patients with a maximum score were excluded from the analyses; they maintained that score throughout the study period.

In the clinical study, intraventricular access device-related infections were observed in two patients. In each case, antibiotics were administered, the intraventricular access device was replaced and the patient continued on cerliponase alfa treatment. Hypotension was reported in 2 (8%) patients, which occurred during or up to 8 hours after cerliponase alfa infusion. Patients did not require alteration in treatment and reactions resolved spontaneously or after IV fluid administration.

Hypersensitivity reactions have been reported in 11 (46%) cerliponase alfa treated patients during the clinical studies. Undesirable or hypersensitivity reactions related to cerliponase alfa treatment, including fever, vomiting, and irritability, may occur during treatment and as late as 24 hours after infusion. Children may receive medication such as antihistamines before cerliponase alfa infusions to reduce the risk of reactions. Serious and severe allergic reactions (anaphylaxis) may occur. If a severe reaction occurs, the infusion is to be stopped immediately and appropriate medical treatment initiated.

The most common adverse reactions (8% or more) are pyrexia, ECG abnormalities, decreased CSF protein, vomiting, seizures, device-related complications, hypersensitivity, increased CSF protein, hematoma, headache, irritability, pleocytosis, device-related infections, bradycardia, feeling jittery, and hypotension.

The Brineura label includes the following contraindications:

- Any sign or symptom of acute or unresolved localized infection on or around the device insertion site (e.g., cellulitis or abscess); or suspected or confirmed CNS infection (e.g., cloudy CSF or positive CSF gram stain, or meningitis);
- Any acute intraventricular access device-related complication (e.g., leakage, extravasation of fluid, or device failure);
- Patients with ventriculoperitoneal shunts.

A phase 2, open-label clinical trial (NCT02678689) aimed to evaluate cerliponase alpha in symptomatic and presymptomatic CLN2 patients less than 18 years of age. The trial enrolled 14 patients ranging in age from 1 to 6 years at baseline, including 8 patients less than 3 years of age. Patients received cerliponase alfa at the recommended dose every 2 weeks by intraventricular infusion for 144 weeks (1 patient withdrew to receive treatment commercially). The mean baseline CLN2 Motor score was 2.3 (standard deviation (SD) 0.83) with a range from 1 to 3. Thirteen of the 14 treated patients were matched with up to 3 natural history comparators on the basis of age within 3 months, equal CLN2 Motor score, and genotype (0, 1, or 2 key mutations). None of the cerliponase alpha treated patients (N=14) had a 2-point decline or score of zero in the Motor scale by Week 169. Among the matched natural history comparators (N=31), 20 subjects (65%) had an unreversed 2-point decline or score of zero by last assessment. The median time to an unreversed 2-point decline in Motor score or score of 0 was 133 weeks among the natural history comparators and was not reached by last assessment (Week 169) in patients treated with cerliponase alpha. In patients below 3 years of age, none (0%) of the cerliponase alpha treated patients (N=8) had a 2-point decline or score of zero in the Motor score by Week 169. Among the 8 treated patients, 7 were matched to 18 untreated patients from the natural history cohort. Among the matched natural history comparators (N=18), 11 subjects (61%) had an unreversed 2-point decline or score of zero in the Motor score by last

assessment. All seven of the treated patients below 3 years of age with a motor score of 3 at baseline remained at a motor score of 3 at the last measured timepoint, which represents grossly normal gait. In this population cerliponase alpha treated patients showed a delay in disease onset.

In July 2024, the FDA approved the supplemental Biologics License Application (sBLA) for cerliponase alpha (Brineura) to slow the loss of ambulation in children of all ages with neuronal ceroid lipofuscinosis type 2 (CLN2 disease). This expanded indication now includes children of all ages with CLN2 disease, regardless of whether they are symptomatic or presymptomatic. The sBLA was supported by data from the phase 2, open-label, multicenter trial (NCT02678689) evaluating Brineura treatment over the span of approximately three years in children aged 1 to 6 years at baseline, including eight children less than 3 years of age who were administered cerliponase alfa according to the recommended dosing regimen based on age for up to 144 weeks. The data showed that intraventricular-administered Brineura slowed the decline in motor function and delayed disease onset in children with CLN2 disease, including those who were under 3 years of age. The safety profile in this population were similar to the known safety profile of the medicine. Moreover, the data demonstrated that early treatment initiation before 3 years of age may result in delaying disease onset.

Niemann-Pick Disease, Type C

According to the National Organization for Rare Disorders (NORD, 2017), Niemann-Pick disease type C (NPC) is a rare progressive genetic disorder characterized by an inability of the body to transport cholesterol and other lipids inside of cells, which leads to abnormal accumulation within various tissues of the body, including brain tissue. Individuals with NPC have mutations in one of two genes, NPC1 or NPC2. Symptoms can present from the perinatal period until late adulthood. NPC affects neurologic and psychiatric

functions, as well as various visceral organs. Symptoms arise at different times and follow independent progression. Visceral symptoms are more typically seen in individuals presenting at a younger age. Neurologic and psychiatric symptoms often occur slowly over time, and thus feature more prominently in individuals presenting in the later age groups. Current treatment is directed toward the specific symptoms apparent in each individual. Miglustat (Zavesca) may be able to slow the progression of neurological symptoms associated with NPC. Miglustat blocks the synthesis of glycosphingolipids, one of the substances that accumulates in the brain of individuals with NPC. Miglustat has been used off-label in the U.S. to treat individuals with NPC.

Pineda et al. (2018) state that miglustat is indicated for the treatment of progressive neurological manifestations in both adults and children. The authors conducted a comprehensive review of published data from studies of cellular neuropathological markers and structural neurological indices in the brain, clinical impairment/disability, specific clinical neurological manifestations, and patient survival. The authors found that "cranial diffusion tensor imaging and magnetic resonance spectroscopy studies have shown reduced levels of choline (a neurodegeneration marker), and choline/N-acetyl aspartate ratio (indicating increased neuronal viability) in the brain during up to 5 years of miglustat therapy, as well as a slowing of reductions in fractional anisotropy (an axonal/myelin integrity marker). A 2-year immunoassay study showed significant reductions in CSF-calbindin during treatment, indicating reduced cerebellar Purkinje cell loss. Magnetic resonance imaging studies have demonstrated a protective effect of miglustat on cerebellar and subcortical structure that correlated with clinical symptom severity. Numerous cohort studies assessing core neurological manifestations (impaired ambulation, manipulation, speech, swallowing, other) using NP-C disability scales indicate neurological stabilization over 2–8 years, with a trend for greater benefits in patients with older (non-infantile) age at neurological onset. A randomized controlled trial and several cohort

studies have reported improvements or stabilization of saccadic eye movements during 1–5 years of therapy. Swallowing was also shown to improve/remain stable during the randomized trial (up to 2 years), as well as in long-term observational cohorts (up to 6 years). A meta-analysis of dysphagia – a potent risk factor for aspiration pneumonia and premature death in NP-C – demonstrated a survival benefit with miglustat due to improved/stabilized swallowing function". The authors concluded that the effects of miglustat on neurological manifestations has been assessed using a range of approaches, with benefits ranging from cellular changes in the brain through to visible clinical improvements and improved survival.

A review in UpToDate "Overview of Niemann-Pick disease" (Patterson, 2020) states that there is no treatment for Niemann-Pick disease that is proven to modify the onset or neurologic progression of the disease, or to prolong lifespan. However, some patients with Niemann-Pick disease type C (NPC) may benefit from treatment with miglustat, which may delay the progression of the neurologic manifestations in children without severe neurologic symptoms at the start of treatment. However, expert consensus guidelines note that miglustat should not be given to patients who have no neurologic manifestations because some remain asymptomatic for long periods of time. Patterson suggests miglustat for patients who have mild to moderate neurologic, psychiatric, or cognitive manifestations. Treatment should be started as soon as any neurologic manifestations emerge. In addition, patients and their families should be informed that the effectiveness of miglustat for NPC is unproven, and that the best attainable outcome of therapy is neurologic stabilization or a slower rate of neurologic disease progression.

Pompe Disease

Infantile Pompe disease (IPD), also known as infantile acid maltase deficiency and type 2 glycogen storage disease, is an autosomal recessive muscle-wasting disorder due to a deficiency of the lysosomal enzyme acid alpha-glucosidase. The deficiency results in accumulation of glycogen in lysosomes and is characterized by progressive cardiomyopathy, skeletal muscle weakness and respiratory insufficiency leading to death in early infancy. Pompe disease is estimated to occur in about 1 in 40,000 live births.

Recombinant human alglucosidase alfa (rhGAA; Myozyme) provides exogenous human lysosomal acid alpha-glucosidase. Once available systemically, alglucosidase alfa mimics endogenous acid alpha-glucosidase. Alglucosidase alfa is indicated for use in members with Pompe disease (GAA deficiency).

On April 28, 2006, the FDA approved alglucosidase alfa, rhGAA (Myozyme), the first treatment for IPD. Alglucosidase alfa had been granted FDA orphan drug status and was approved under a priority review. Myozyme is indicated for the treatment of Pompe disease and has been shown to improve ventilator-free survival in members with infantile-onset (initiated in children 1 month- 3.5 years) Pompe disease, use of Myozyme in members with other forms of Pompe disease has not been adequately studied to assure safety and efficacy.

Alglucosidase alpha (Myozyme) is available as a 50mg vial for reconstitution. Alglucosidase alfa (Genzyme Corp, Cambridge, MA) is administered by intravenous infusion every 2 weeks. The recommended dosage is 20 mg/kg body weight administered over approximately 4 hours as an intravenous infusion. The total volume of infusion is determined by the member's body weight.

Initiation of alglucosidase alfa therapy in the earliest stages of the disease process is associated with the most clinical benefit. Infusions should be administered in a stepwise manner using an

infusion pump. The initial infusion rate should be no more than 1mg/kg/hr. The infusion rate may be increased by 2 mg/kg/hr every 30 minutes, after member tolerance to the infusion rate is established, until a maximum rate of 7 mg/kg/hr is reached. Vital signs should be obtained at the end of each step.

The safety and efficacy of alglucosidase alfa were assessed in 2 separate clinical trials in patients (n = 39) with infantile-onset Pompe disease ranging in age from 1 month to 3.5 years at the time of the first infusion. Patient survival without needing invasive ventilatory support was substantially greater in the alglucosidase alfa-treated infants than would be expected considering the high mortality rate of untreated patients of similar age and disease severity. According to the FDA, the drug's safety and effectiveness in other forms of Pompe disease have not been adequately studied.

A clinical study of the efficacy of aglucosidase alpha in persons with late-onset Pompe disease found statistically significant but modest effects (van der Ploeg et al, 2010). A total of 90 patients who were 8 years of age or older, ambulatory, and free of invasive ventilation were randomly assigned to receive bi-weekly intravenous alglucosidase alfa (20 mg/kg) or placebo for 78 weeks at 8 centers in the U.S. and Europe. The 2 primary end points were distance walked during a 6-min walk test and percentage of predicted forced vital capacity (FVC). At 78 weeks, the estimated mean changes from baseline in the primary end points favored alglucosidase alfa (an increase of 28.1 +/- 13.1 meters on the 6-min walk test and an absolute increase of 3.4 +/- 1.2 percentage points in FVC; p = 0.03 and p = 0.006, respectively). The authors concluded that their data indicated that alglucosidase alfa treatment, as compared with placebo, had a positive, if modest, effect on walking distance and pulmonary function in patients with late-onset Pompe's disease and may stabilize proximal limb and respiratory muscle strength. The authors noted that the study had a number of limitations. They noted that, although 90 patients is a large population for a clinical

trial designed to study an orphan disease, the number is relatively small when the goal is to judge the progression of a clinically heterogeneous disease. Before the start of this trial, no longitudinal data were available on changes in the 6-min walk test over time in patients with untreated Pompe's disease, and the mean decline in the distance walked was minimal in the patients in this study who received placebo. "Longer follow-up will be needed to confirm our results, given the variable presentation and rate of deterioration among the patients in our study and the possible effect of the degree of muscle destruction at baseline on their response to treatment" (van der Ploeg et al, 2010).

In this study, similar proportions of patients in the 2 groups had adverse events, serious adverse events, and infusion-associated reactions; events that occurred only in patients who received the active study drug included anaphylactic reactions and infusion-associated reactions of urticaria, flushing, hyperhidrosis, chest discomfort, vomiting, and increased blood pressure (each of which occurred in 5 to 8 % of the patients) (van der Ploeg et al, 2010). The most serious adverse reactions reported with alglucosidase alfa were heart and lung failure and allergic shock. Most common reactions included pneumonia, respiratory failure and distress, infections, and fever. A black box warning is included in the Myozyme label to warn about the possibility of life-threatening allergic reactions.

Late-onset glycogen storage disease type 2 (GSD2)/Pompe disease is a progressive multi-system disease evoked by a deficiency of lysosomal acid alpha-glucosidase (GAA) activity. It is characterized by respiratory and skeletal muscle weakness and atrophy, resulting in functional disability and reduced life span. Since 2006, alglucosidase alfa has been licensed as a treatment in all types of GSD2/Pompe disease. Strothotte et al (2010) presented an open-label, investigator-initiated observational study of alglucosidase alfa ERT in 44 late-onset GSD2 patients with various stages of disease severity. Alglucosidase alfa was given

intravenously at the standard dose of 20 mg/kg every other week. Assessments included serial arm function tests (AFT), Walton Gardner Medwin scale (WGMS), timed 10-m walk tests, 4-stair climb tests, modified Gowers' maneuvers, 6-min walk tests, MRC sum score, FVC, creatine kinase (CK) levels and SF-36 self-reporting questionnaires. All tests were performed at baseline and every 3 months for 12 months of ERT. These researchers found significant changes from baseline in the modified Gowers' test, the CK levels and the 6-min walk test (341 +/- 149.49 m, median 342.25 m at baseline; 393 +/- 156.98 m; median 411.50 m at end point; $p = 0.026$), while all other tests were unchanged. Enzyme replacement therapy over 12 months revealed minor allergic reactions in 10 % of the patients. No serious adverse events occurred. None of the patients died or required de novo ventilation. The authors stated that the clinical outcome data imply stabilization of neuromuscular deficits over 1 year with mild functional improvement.

On March 25, 2010, the FDA approved alglucosidase alfa (Lumizyme) for patients aged 8 years and older with late-onset (non-infantile) Pompe disease. Lumizyme is believed to work by replacing the deficient GAA, thereby reducing the accumulated glycogen in heart and skeletal muscle cells. On August 1, 2014, the U.S. Food and Drug Administration (FDA) approved the supplement to expand the indication for Lumizyme to all Pompe patients. The FDA reviewed newly available information and determined that Lumizyme and Myozyme are chemically and biochemically comparable. Consequently, the safety and effectiveness of Lumizyme and Myozyme are expected to be comparable. In addition, a single-center clinical study of 18 infantile-onset Pompe disease patients, aged 0.2 to 5.8 months at the time of first infusion, provided further support that infantile-onset patients treated with Lumizyme will have a similar improvement in ventilator-free survival as those treated with Myozyme.

Currently, the only other treatment for Pompe disease available in the U.S. is Myozyme, which is also manufactured by Genzyme. Myozyme has been in short supply due to limited manufacturing capacity. The manufacturer reserved Myozyme to treat infants and children with Pompe disease because younger patients generally have a much more aggressive form of the disease. Some adult patients in the U.S. received Lumizyme under a temporary access program. The approval of Lumizyme will ensure that treatment is available for all U.S. adult Pompe patients in need of treatment. Lumizyme's safety and effectiveness have not been evaluated in patients with infantile-onset Pompe disease or in patients aged 8 years and younger with late-onset disease. These patients should be treated with Myozyme, not Lumizyme.

Both Myozyme and Lumizyme (alglucosidase alfa) are made by the same manufacturer. Once Lumizyme (alglucosidase alfa) became approved for patients with Pompe disease, Myozyme production was discontinued and all patients were switched to Lumizyme.

Recombinant human alglucosidase alfa (rhGAA; Lumizyme) is a hydrolytic lysosomal glycogen-specific enzyme that provides exogenous human lysosomal acid alphaglucohydrolase. Alglucosidase alfa is used primarily in members with Pompe disease, which is an inherited disorder of glycogen metabolism caused by the relative or absolute absence of acid alpha-glucosidase. Once available systemically, alglucosidase alfa mimics endogenous acid alpha-glucosidase.

Alglucosidase alpha (Lumizyme) is available as a 50mg vial for reconstitution. The recommended dosage of alglucosidase alpha (Lumizyme) is 20 mg/kg body weight administered every 2 weeks as an intravenous infusion. The total volume of infusion is determined by the member's body weight and should be administered over approximately 4 hours.

Initiation of alglucosidase alfa therapy in the earliest stages of the disease process is associated with the most clinical benefit.

Infusions should be administered in a step-wise manner using an infusion pump. The initial infusion rate should be no more than 1mg/kg/hr. The infusion rate may be increased by 2 mg/kg/hr every 30 minutes, after member tolerance to the infusion rate is established, until a maximum rate of 7 mg/kg/hr is reached. Vital signs should be obtained at the end of each step. If the member is stable, Lumizyme may be administered at the maximum rate of 7 mg/kg/hr until the infusion is completed. The infusion rate may be slowed or temporarily stopped in the event of infusion reactions.

Hypersensitivity reactions, including anaphylaxis, immune-mediated reactions, and risk of acute cardiorespiratory failure have been observed in some patients during and after treatment with alglucosidase alfa. Healthcare providers are to ensure that appropriate medical support measures, including cardiopulmonary resuscitation equipment, are readily available. If anaphylaxis or severe hypersensitivity reactions occur, the infusion should immediately be discontinued and appropriate medical treatment initiated.

Patients should be monitored for development of systemic immune-mediated reactions involving skin and other organs.

Patients with compromised cardiac or respiratory function may be at risk of acute cardiorespiratory failure. Caution should be exercised when administering alglucosidase alfa to patients susceptible to fluid volume overload. Appropriate medical support and monitoring measures should be available during infusion.

There is a risk of cardiac arrhythmia and sudden cardiac death during general anesthesia for central venous catheter placement; caution should be used when administering general anesthesia for the placement of a central venous catheter intended for alglucosidase alfa infusion.

Cupler and colleagues (2012) proposed consensus-based treatment and management recommendations for late-onset Pompe disease. A systematic review of the literature by a panel of specialists with expertise in Pompe disease was undertaken. A multi-disciplinary team should be involved to properly treat the pulmonary, neuromuscular, orthopedic, and gastro-intestinal elements of late-onset Pompe disease. Pre-symptomatic patients with subtle objective signs of Pompe disease (and patients symptomatic at diagnosis) should begin treatment with ERT immediately; pre-symptomatic patients without symptoms or signs should be observed without use of ERT. After 1 year of ERT, patients' condition should be re-evaluated to determine whether ERT should be continued.

On August 6, 2021, the U.S. FDA approved avalglucosidase alfa-ngpt, brand Nexviazyme (Genzyme Corporation), for the treatment of patients one year of age and older with late-onset Pompe disease (LOPD). Nexviazyme is an enzyme replacement therapy (ERT) designed to specifically target the mannose-6-phosphate (M6P) receptor, the key pathway for cellular uptake of enzyme replacement therapy in Pompe disease. FDA approval is based on results from the COMET study that compared Nexviazyme to avalglucosidase alfa in LOPD.

The COMET study (NCT02782741) was a randomized, double-blinded, multinational, multicenter, phase 3 trial comparing the efficacy and safety of Nexviazyme to alglucosidase alfa in 100 treatment-naïve patients with LOPD. Patients were randomized in a 1:1 ratio based on baseline forced vital capacity (FVC), gender, age, and country to receive 20 mg/kg of Nexviazyme or alglucosidase alfa administered intravenously once every two weeks for 49 weeks. The trial included an open-label, long-term, follow-up phase of up to 5 years, in which patients in the alglucosidase alfa arm were switched to Nexviazyme treatment. Of the 100 randomized patients, the baseline median age was 49 years old (range from 16 to 78), median length of time

since diagnosis was 6.9 months, mean age at diagnosis was 46.4 years old (range from 11 to 78), mean FVC (% predicted) at baseline was 62.1% (range from 32 to 85%), and mean 6-minute walk test (6MWT) at baseline was 388.9 meters (range from 118 to 630 meters). The primary endpoint was the change in FVC (% predicted) in the upright position from baseline to Week 49. At Week 49, the least squares (LS) mean change in FVC (% predicted) for patients treated with Nexviazyme and alglucosidase alfa was 2.9% and 0.5%, respectively. The estimated treatment difference was 2.4% (95% CI: -0.1, 5.0) favoring Nexviazyme. In summary, when compared to baseline, patients treated with Nexviazyme had a 2.9-point improvement (SE=0.9) in FVC percent-predicted at Week 49, the study's primary endpoint. Patients treated with Nexviazyme had a 2.4-point greater improvement in FVC percent-predicted compared to patients treated with alglucosidase alfa at Week 49 meeting the measurement of non-inferiority ($p=0.0074$). Statistical superiority of Nexviazyme over alglucosidase alfa was not achieved ($p=0.06$). A key secondary endpoint in the trial measured functional endurance with the 6MWT. When compared to baseline, patients treated with Nexviazyme walked 32.2 meters farther (SE=9.9) at Week 49. Patients treated with Nexviazyme walked 30 meters farther than patients treated with alglucosidase alfa at Week 49. Per the hierarchy of the study protocol, formal statistical testing for all secondary endpoints was not conducted (Genzyme, 2021; Sanofi, 2021).

Nexviazyme carries a boxed warning for severe hypersensitivity reactions, infusion-associated reactions, and risk of acute cardiorespiratory failure in susceptible patients. The most common adverse reactions (5% or more) in clinical trials were hypersensitivity reactions and included: anaphylaxis, rash, pyrexia, flushing/feeling hot, urticaria, headache, hyperhidrosis, nausea, cough, decreased oxygen saturation, tachycardia, tachypnea, chest

discomfort, dizziness, muscle twitching, agitation, cyanosis, erythema, hypertension/increased blood pressure, pallor, rigors, tremor, vomiting, fatigue, and myalgia.

In September 2023, the FDA approved Pombiliti (cipaglucosidase alfa-atga) infusion in combination with Opfolda (miglustat) 65mg capsules for the treatment of adults with late-onset Pompe disease (LOPD) weighing greater than or equal to 40 kg and who are not improving on their current enzyme replacement therapy (ERT). Pombiliti is a recombinant human lysosomal acid alpha-glucosidase enzyme (rhGAA) which replaces the missing enzyme acid alpha-glucosidase (cipaglucosidase alfa-agta). When absorbed into the muscle cell, Pombiliti gets converted to GAA and is then able to break down glycogen, reducing glycogen toxic levels. Opfolda is an enzyme stabilizer designed to stabilize the enzyme in the blood.

FDA approval of Pombiliti is based on clinical data observed from a randomized, double-blind, active-controlled, international, multi-center, phase 3 study (PROPEL). The ERT-experienced patients treated with Pombiliti in combination with Opfolda showed a numerically favorable change in sitting forced vital capacity (FVC) and 6-minute walk distance (6MWD) from baseline at Week 52. Results showed that at Week 52, patients treated with Pombiliti plus Opfolda (i) experienced a mean change in sitting FVC of -1.1% compared with -3.3% for alglucosidase alfa-treated patients (estimated treatment difference, 2.3% [95% CI, 0.02-4.62]), (ii) among ERT-experienced patients, the Pombiliti plus Opfolda arm showed a numerically favorable change in sitting FVC at week 52 (nominal $p = .006$), (iii) patients walked on average 21 meters farther from baseline compared with 8 meters for those treated with alglucosidase alfa (estimated treatment difference was 14 meters [95% CI, -1, 28]), and (iv) among ERT-experienced patients, the Pombiliti plus Opfolda arm experienced a numerically favorable change in 6MWD at week 52 (nominal $p = .047$) (Amicus Therapeutics US, 2023).

In the PROPEL study, Schoser and colleagues (2021) evaluated the safety and efficacy of cipaglucosidase alfa plus miglustat in adults with late-onset Pompe disease who were still ambulatory, able to breathe without ventilation support, had an enzyme assay demonstrating a deficiency of acid alpha-glucosidase enzyme activity or diagnosis confirmed by genetic testing, weight greater than equal to 40 kg, and who had either been receiving alglucosidase alfa for at least 2 years or were ERT-naïve. Patients 18 years and older were randomized 2:1 to intravenous cipaglucosidase alfa (20 mg/kg) plus oral miglustat or to intravenous alglucosidase alfa (20 mg/kg) plus oral placebo once every 2 weeks for 52 weeks. Patients, investigators, and outcome assessors were masked to treatment assignment. The primary endpoint was change from baseline to week 52 in 6MWD, assessed using a mixed-effect model for repeated measures analysis for comparison of superiority in the intention-to-treat population (all patients who received at least one dose of study drug). Of the patients screened for eligibility, 125 were enrolled and randomly assigned to receive cipaglucosidase alfa plus miglustat (n=85) or alglucosidase alfa plus placebo (n=40). Two patients in the alglucosidase alfa plus placebo group did not receive any dose due to absence of genotype confirmation of late-onset Pompe disease and were excluded from analysis. Six patients discontinued (one in the alglucosidase alfa plus placebo group, five in the cipaglucosidase alfa plus miglustat group), and 117 completed the study. At week 52, mean change from baseline in 6MWD was 20·8 m (SE 4·6) in the cipaglucosidase alfa plus miglustat group versus 7·2 m (6·6) in the alglucosidase alfa plus placebo group using last observation carried forward (between-group difference 13·6 m [95% CI -2·8 to 29·9]). 118 (96%) of 123 patients experienced at least one treatment-emergent adverse event during the study; the incidence was similar between the cipaglucosidase alfa plus miglustat group (n=81 [95%]) and the alglucosidase alfa plus placebo group (n=37 [97%]). The most frequently reported treatment-emergent adverse events were fall (25 [29%] patients in the cipaglucosidase alfa plus miglustat group vs 15 [39%] in the

alglucosidase alfa plus placebo group), headache (20 [24%] vs 9 [24%]), nasopharyngitis (19 [22%] vs 3 [8%]), myalgia (14 [16%] vs 5 [13%]), and arthralgia (13 [15%] vs 5 [13%]). The investigators noted that 12 serious adverse events occurred in eight patients in the cipaglucosidase alfa plus miglustat group; with one event (anaphylaxis) deemed related to the study drug. One serious adverse event (stroke) occurred in the alglucosidase alfa plus placebo group, which was deemed unrelated to study drug. There were no deaths. The investigators concluded that cipaglucosidase alfa plus miglustat did not achieve statistical superiority to alglucosidase alfa plus placebo for improving 6MWD in their overall population of patients with late-onset Pompe disease. Further studies should investigate the longer-term safety and efficacy of cipaglucosidase alfa plus miglustat and whether this investigational two-component therapy might provide benefits, particularly in respiratory function and in patients who have been receiving enzyme replacement therapy for more than 2 years, as suggested by their secondary and subgroup analyses.

Pombiliti carries a boxed warning for severe hypersensitivity reactions, infusion-associated reactions, and risk of acute cardiorespiratory failure in susceptible patients. Patients susceptible to fluid volume overload, or those with acute underlying respiratory illness or compromised cardiac or respiratory function, may be at risk of serious exacerbation of their cardiac or respiratory status during infusion therapy. Additional warnings and precautions include risk of embryo-fetal toxicity and risk associated with Opfolda. The most common adverse reaction (5% or more) include headache, diarrhea, fatigue, nausea, abdominal pain, and pyrexia. Breastfeeding is not recommended.

Miscellaneous Interventions

Hematopoietic Stem Cell Therapy

Biffi (2017) stated that lysosomal storage disorders (LSDs) are a broad class of monogenic diseases with an overall incidence of 1:7,000 newborns, due to the defective activity of one or more lysosomal hydrolases or related proteins resulting in storage of undegraded substrates in the lysosomes. The over 40 different known LSDs share a life-threatening nature, but they are present with extremely variable clinical manifestations, determined by the characteristics and tissue distribution of the material accumulating due to the lysosomal dysfunction. The majority of LSDs lack a curative treatment. This is particularly true for LSDs severely affecting the central nervous system (CNS). Based on current pre-clinical and clinical evidences, among other treatment modalities, hematopoietic stem cell therapy (HSCT) could potentially result in robust therapeutic benefit for LSD patients, with particular indication for those characterized by severe brain damage. The authors concluded that optimization of current approaches and technology, as well as implementation of clinical trials for novel indications, and prolonged and more extensive follow-up of the already treated patients will allow translating this promise into new medicinal products.

Azario and associates (2017) noted that umbilical cord blood (UCB) is a promising source of stem cells to use in early HSCT approaches for several genetic diseases that can be diagnosed at birth. Mucopolysaccharidosis type I (MPS-I) is a progressive multi-system disorder caused by deficiency of lysosomal enzyme α -L-iduronidase, and patients treated with allogeneic HSCT at the onset have improved outcome, suggesting to administer such therapy as early as possible. Given that the best characterized MPS-I murine model is an immunocompetent mouse, these researchers developed a transplantation system based on murine UCB. With the final aim of testing the therapeutic efficacy of UCB in MPS-I mice transplanted at birth, these investigators first defined the features of murine UCB cells and demonstrated that they were capable of multi-lineage hematopoietic re-population of myelo-ablated adult mice similarly to bone marrow cells. They then

assessed the effectiveness of murine UCB cells transplantation in busulfan-conditioned newborn MPS-I mice; 20 weeks after treatment, iduronidase activity was increased in visceral organs of MPS-I animals, glycosaminoglycans storage was reduced, and skeletal phenotype was ameliorated. The authors concluded that this study explored a potential therapy for MPS-I at a very early stage in life and represents a novel model to test UCB-based transplantation approaches for various diseases.

Somaraju and Tadepalli (2017) stated that Gaucher disease is the most common LSD caused by a deficiency of the enzyme glucocerebrosidase. Current treatment of the disease involves a choice from ERT, substrate reduction therapy and HSCT.

Hematopoietic stem cell therapy is a high risk procedure with possible long-term benefits in the regression of skeletal and neurological changes in people with Gaucher disease. This is an update of a previously published Cochrane Review. These investigators determined the role of HSCT in people with Gaucher disease in relation to: mortality risk associated with the procedure; efficacy in modifying the course of the disease; and arrest or regression of neurological manifestations in neuronopathic forms (types 2 and 3). They searched the Cochrane Cystic Fibrosis and Genetic Disorders Group Inborn Errors of Metabolism Trials Register which comprises of references identified from comprehensive electronic database searches and hand-searches of relevant journals and abstract books of conference proceedings. Date of the most recent search of the Group's Hemoglobinopathies Trials Register: January 19, 2017. These investigators also searched the websites: www.clinicaltrials.gov; WHO International Clinical Trials Registry Platform portal and www.genzymeclinicalresearch.com. Date of most recent search of these sites: 02 March 2017. All randomized, quasi-randomized and controlled clinical trials comparing SCT with ERT, substrate reduction therapy, symptomatic treatment or no treatment in people with Gaucher disease of all ages were selected for analysis. The authors independently assessed trials for inclusion, however, no

relevant trials were identified. A total of 32 trials were identified by the searches; however, these were not suitable for inclusion in the review. The authors concluded that HSCT is a form of treatment that offers the potential of permanent cure. However, there were no clinical trials that have assessed the safety and efficacy of this treatment in comparison to other conservative measures (ERT, substrate reduction therapy) now in use. There were no trials included in the review and these researchers had not identified any relevant trials up to March 2017.

Wright and colleagues (2017) described long-term outcomes of children with early-infantile Krabbe disease who underwent HSCT in the first 7 weeks of life. In this prospective longitudinal study, evaluations were performed at baseline and follow-up included brain imaging, neuro-diagnostic tests, and neuro-behavioral evaluations. Of the 18 patients in this study (11 girls, 7 boys; mean follow-up of 9.5 years, range of 4 to 15), 5 died (3 of peri-transplant complications, 1 of a surgical complication unrelated to Krabbe disease, 1 of disease progression); 1 of the surviving patients had normal cognitive function and 10 continue to develop cognitive skills at a slightly slower rate than normal. All surviving patients continue to gain receptive language skills, with 7 falling within the normal range; 10 patients received speech therapy, and 2 of these patients required augmentative communication devices. Gross motor development varies widely, but 3 patients could walk independently, and 7 walked with assistive devices. Spasticity ranged from mild to severe, and 12 patients wore orthotics. Fine motor skills were generally preserved. Brain myelination and atrophy stabilized in 8 patients, improved in 4 patients, and worsened in 1 patient. Nerve conduction velocities initially improved but continued to be abnormal in most patients. The authors concluded that surviving patients functioned at a much higher level than untreated children or symptomatic children who underwent HSCT. They stated that these results showed that early HSCT changed the natural history of this disease by improving both lifespan and functional abilities. Moreover, the authors stated that

continued follow-up studies are needed to monitor the long-term outcomes of treated patients as they age into adolescence and young adulthood. It will also be critical to identify a pre-symptomatic diagnostic biomarker that will enable clinicians to confidently follow babies at moderate and high risk for Krabbe disease identified through newborn screening programs and those treated with HSCT or any other future therapies. This study provided Class IV evidence that for children with early-infantile Krabbe disease, early HSCT improved lifespan and functional abilities.

In Utero Enzyme-Replacement Therapy for Infantile-Onset Pompe Disease

Cohen et al (2022) stated that patients with early-onset lysosomal storage diseases are ideal candidates for pre-natal therapy because organ damage starts in-utero. These investigators reported the safety and effectiveness of in-utero ERT in a fetus with CRIM (cross-reactive immunologic material)-negative infantile-onset Pompe disease. The family history was positive for infantile-onset Pompe disease with cardiomyopathy in 2 previously affected deceased siblings. After receiving in-utero ERT and standard post-natal therapy, the patient exhibited normal cardiac and age-appropriate motor function postnatally, was meeting developmental milestones, had normal biomarker levels, and was feeding and growing well at 13 months of age. Moreover, these researchers stated that similar to any fetal intervention, this procedure is associated with a risk of pre-term delivery. Non-directive counseling for parents of affected pregnancies is paramount to ensure informed choices regarding the risk-benefit profile of this new therapeutic approach. These investigators noted that their phase-I clinical trial of in-utero ERT for fetuses with genetically confirmed early-onset lysosomal storage diseases will enable the collection of additional safety and effectiveness data in a larger cohort.

In an editorial on the afore-mentioned study by Cohen et al (2022), van der Ploeg (2022) stated that an issue that needs to be clarified is whether initiation of therapy at 24 weeks of gestation sufficiently precedes the formation of the blood-brain barrier to prevent the neurologic phenotypes observed in many lysosomal storage disorders. Unfortunately, the short half-life of the lysosomal (replacement) enzyme weighs against the likelihood that neurologic phenotypes can be entirely prevented. The editorialist stated that further follow-up of this patient will be important; Cohen and colleagues are conducting a phase-I clinical trial of in-utero ERT in 10 fetuses with a genetically diagnosed lysosomal storage disorder. The outcomes of this phase-I clinical trial are eagerly awaited; in the meantime, the developmental course of this patient is encouraging.

Furthermore, an UpToDate review on “Lysosomal acid alpha-glucosidase deficiency (Pompe disease, glycogen storage disease II, acid maltase deficiency)” (Hahn, 2023) does not mention in-utero ERT as a management / therapeutic option.

Measurement of Anti-rhGAA Antibodies as Determinant of Treatment Outcome in Adults with Late-Onset Pompe Disease

Ditters et al (2023) noted that Pompe disease is a lysosomal storage disease characterized by skeletal and respiratory muscle weakness. Since 2006, ERT with alglucosidase alfa has been available. ERT significantly improves the prognosis of patients with Pompe disease. The effect of high antibody titers on treatment response in adults with LOPD remains unclear, but may contribute to inter-patient variation. In a systematic review on this subject, these investigators carried out a literature search in Embase, Medline Ovid, Web of Science, Psych Info Ovid, Cochrane (Clinical Trials only), and Google Scholar (random top-200). Studies were included if they involved adults with LOPD treated with alglucosidase alfa and mentioned anti-rhGAA antibodies or antibody titers. Furthermore, studies mentioning dosages different

from the standard recommended dosage were included. The literature search retrieved 2,562 publications, and 17 fulfilled the selection criteria, describing 443 cases -- 7 studies reported on anti-rhGAA antibody titers on a group level, with the percentage of patients with a high titer as defined in the included articles ranging from 0 % to 33 %; 6 studies reported on the effect of anti-rhGAA antibody titer on clinical course, and 4 found no correlation. Two studies reported a negative effect on treatment. The 1st study found a greater improvement in Medical Research Council (MRC) score in patients with no detectable antibody titer. In the 2nd study, a patient discontinued ERT due to a declining neuromuscular state as a result of high anti-rhGAA antibody titers; 7 studies reported on 17 individual patients with a high antibody titer (range of 1:12,800 to 1:3,906,250). In only 2 cases were high-sustained neutralizing antibodies reported to interfere with treatment effectiveness. The authors concluded that no clear effect of anti-rhGAA IgG antibodies on therapeutic response could be established for the majority of LOPD patients with a high antibody titer. In a minority of patients, a clinical decline related to (possible) interference of anti-rhGAA antibodies was described. These researchers stated that further prospective, long-term studies are needed to evaluate this relation. They stated that for future investigations, these researchers advised to standardize the definition of a high antibody titer (greater than or equal to 31,250) and of a high sustained antibody titer (high antibody titer on 2 or more occasions at or beyond 6 months of ERT) to improve comparability; to perform regular, standardized follow-up of antibody titer (with collection of samples before ERT infusion), which will provide insight into the development and course of antibody titer over time and prevent sampling bias; and to describe patient characteristics and standardize clinical follow-up to enable assessment of the effect of antibody formation on outcome of ERT in individual patients.

The authors stated that this analysis had several drawbacks. First, different definitions of a high antibody titer as well as a high sustained antibody titer were given, making it difficult to compare

study outcomes in this regard. The definition of a high antibody titer ranged from 1:12,800 to greater than or equal to 1:51,200, which may be partially dependent on the titration steps used. Most studies did not specify what the source was for the reference values for (highly sustained) antibodies. It has been demonstrated in the serum of classic-infantile Pompe patients that there is probably no effect on enzyme activity or uptake of alglucosidase alfa with an antibody titer of less than or equal to 1:6250.

Alglucosidase alfa activity in the medium was not inhibited until titers of 1:31,250 were reached, indicating that it was only useful to measure neutralizing antibodies for a titer of greater than or equal to 1:31,250. This same study also arithmetically estimated that, whereas titers of 1:31,250 may counteract ERT at a dosage of 20 mg/kg, titers above 1:60,000 are expected to counteract ERT when a dosage of 40 mg/kg is administered. These estimates are consistent with the suggested cut-off value for a high titer of 1:51,200, and demonstrated that, to interpret the clinical effect of anti-rhGAA antibodies, the dosage of ERT must be known as well. These researchers suggested using the following definitions of a high and high sustained antibody titer: a high titer means a titer of greater than or equal to 31,250, and a high sustained antibody titer means a titer of greater than or equal to 31,250 on 2 or more occasions at or beyond 6 months of ERT. Second, the level of evidence of the included studies was variable, with 5 out of 17 included studies being case reports, and the outcome of interest (antibody titer and ERT dosage) was often not the main outcome of the paper. As some patients were reported in multiple publications – these investigators were aware of at least 1 patient for whom this was the case -- the number of unique patients described in this review was lower than 443. Multiple studies included patients with different phenotypes that contributed to a larger clinical heterogeneity than intended for this study. Third, due to the heterogeneity of clinical outcome parameters, the different definitions of high (sustained) antibody titers, different time-points at which antibody titers were measured, as well as limited data regarding dosing, these researchers could not analyze antibodies

over time, nor could a meta-analysis be carried out. These investigators resolved this by using the definition of a high (sustained) titer as used in the original publication and by looking at the highest antibody titer reported per patient. Fourth, there might be a selection bias in that patients with high antibody titers or infusion-associated reactions (IARs) were more likely to be reported, and that follow-up was not always continuous over time and varied between studies.

Measurement of Plasma Lysosphingolipids and Oxysterols for Screening Lipid Storage Disorders

Voorink-Moret and colleagues (2018) noted that in patients suspected of a lipid storage disorder (sphingolipidoses, lipidoses), confirmation of the diagnosis relies predominantly on the measurement of specific enzymatic activities and genetic studies. New UPLC-MS/MS methods have been developed to measure lysosphingolipids and oxysterols, which, combined with chitotriosidase activity may represent a rapid first-tier screening for lipid storage disorders. A lysosphingolipid panel consisting of lysoglobotriaosylceramide (LysoGb3), lysohexosylceramide (LysoHexCer: both lysoglucosylceramide and lysogalactosylceramide), lysosphingomyelin (LysoSM) and its carboxylated analogue lysosphingomyelin-509 (LysoSM-509) was measured in control subjects and plasma samples of predominantly untreated patients affected with lipid storage disorders (n = 74). In addition, the oxysterols cholestane-3 β ,5 α ,6 β -triol and 7-ketocholesterol were measured in a subset of these patients (n = 36) as well as chitotriosidase activity (n = 43). A systematic review of the literature was performed to assess the usefulness of these biochemical markers. Specific elevations of metabolites, i.e., without overlap between controls and other lipid storage disorders, were found for several lysosomal storage diseases: increased LysoSM levels in acid sphingomyelinase deficiency (Niemann-Pick disease type A/B), LysoGb3 levels in males with classical phenotype Fabry disease and LysoHexCer (i.e.,

lysoglucosylceramide/lysogalactosylceramide) in Gaucher and Krabbe diseases. While elevated levels of LysoSM-509 and cholestane-3 β ,5 α ,6 β -triol did not discriminate between Niemann Pick disease type C and acid sphingomyelinase deficiency, LysoSM-509/LysoSM ratio was specifically elevated in Niemann-Pick disease type C. In Gaucher disease type I, mild increases in several lysosphingolipids were found including LysoGb3 with levels in the range of non-classical Fabry males and females. Chitotriosidase showed specific elevations in symptomatic Gaucher disease, and was mildly elevated in all other lipid storage disorders. Review of the literature identified 44 publications. Most findings were in line with this study cohort. Several moderate elevations of biochemical markers were found across a wide range of other, mainly inherited metabolic, diseases. The authors concluded that measurement in plasma of lysosphingolipids and oxysterols by UPLC-MS/MS in combination with activity of chitotriosidase provided a useful first-tier screening of patients suspected of lipid storage disease. The LysoSM-509/LysoSM ratio is a promising parameter in Niemann-Pick disease type C. Moreover, they stated that further studies in larger groups of untreated patients and controls are needed to improve the specificity of the findings.

Furthermore, an UpToDate review on "Screening for lipid disorders in adults" (Vijan, 2020) does not mention measurements of plasma lysosphingolipids and oxysterols as screening tools.

Metallothioneins

Cavalca and colleagues (2018) noted that LSDs are a broad class of inherited metabolic diseases caused by the defective activity of lysosomal enzymes; CNS manifestations are present in roughly 50 % of LSD patients and represent an unmet medical need for them.

These researchers examined the therapeutic potential of metallothioneins (MTs), a newly identified family of proteins with reported neuroprotective roles, in the murine models of 2 LSDs with

CNS involvement. MT-1 over-expressing transgenic mice (MTtg) were crossed with the murine models of Batten and Krabbe diseases. Changes in the survival and manifestations of the disease in the MTtg setting were assessed. In addition, these researchers analyzed the therapeutic effects of MT-1 CNS gene delivery in one of these LSD models. Constitutive expression of MT-1 exerted favorable phenotypic effects in both LSD models.

MT-LSD mice showed a 5 % to 10 % increase in survival and slower disease progression as compared to not-transgenic LSD mice. Rescue of Purkinje cells from degeneration and apoptosis was also observed in the MT-LSD models. This phenotypic amelioration was accompanied by a modulation of the disease-associated activated inflammatory microglia phenotype, and by a reduction of oxidative stress. Importantly, for the clinical translation of these findings, the very same effects were obtained when MTs were delivered to brains by systemic AAV gene transfer. The authors concluded that MTs can be considered novel therapeutic agents (and targets) in LSDs and potentiate the effects of approaches aiming at correction of the disease-causing enzyme deficiency in the CNS.

Substrate Deprivation Therapy

Derrick-Roberts and colleagues (2017) stated that MPS I is the most common form of the MPS group of genetic diseases; it results from a deficiency in the lysosomal enzyme α -L-iduronidase, leading to accumulation of un-degraded heparan and dermatan sulphate glycosaminoglycan (GAG) chains in patient cells. Children with MPS suffer from multiple organ failure and die in their teens to early 20s. In particular, MPS I children also suffer from profound mental retardation and skeletal disease that restricts growth and movement. Neither neurological nor skeletal disease is adequately treated by current therapy approaches. To overcome these barriers to effective therapy these researchers have developed and tested a treatment called substrate deprivation therapy (SDT). MPS I knockout mice were treated with weekly intravenous injections of 1

mg/kg rhodamine B for 6 months to assess the efficacy of SDT.

Mice were assessed using biochemistry, micro-CT and a battery of behavioral tests to determine the outcome of treatment. A reduction in female body-weight gain was observed with the treatment as well as a decrease in lung GAG. Behavioral studies showed slight improvements in inverted grid and significant improvements in learning ability for female MPS I mice treated with rhodamine B. Skeletal disease also improved with a reduction in bone mineral volume observed. The authors concluded that rhodamine B is safe to administer to MPS I knockout mice where it had an effect on improving aspects of neurological and skeletal disease symptoms and may therefore provide a potential therapy or adjunct therapy for MPS I patients.

Appendix

Table: Spectrum of mucopolysaccharidosis I (MPS I)

Hurler	Hurler-Scheie	Scheie
■ Most severe form of MPS I ■ More progressive ■ Severe intellectual disability, developmental delay ■ Progressive loss of vision and hearing ■ Skeletal abnormalities (dysostosis multiplex) ■ Hepatosplenomegaly ■ Severe respiratory disease	■ Intermediate in severity ■ Less common than Hurler ■ Progresses less rapidly than Hurler syndrome ■ Little or no intellectual defect ■ Respiratory disease ■ Obstructive airway disease ■ Cardiovascular disease	■ Least severe form of MPS I ■ Normal intelligence ■ Less progressive physical problems ■ Corneal clouding ■ Joint stiffness ■ Carpal

■ Obstructive airway disease	■ Joint stiffness/contractures	tunnel syndrome
■ Cardiovascular disease	■ Skeletal abnormalities	■ Aortic valve heart disease
■ Death before age 10 years	■ Decreased visual acuity	■ Death in middle decades
	■ Death in teens and 20s	

Source: Hahn, 2019

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